

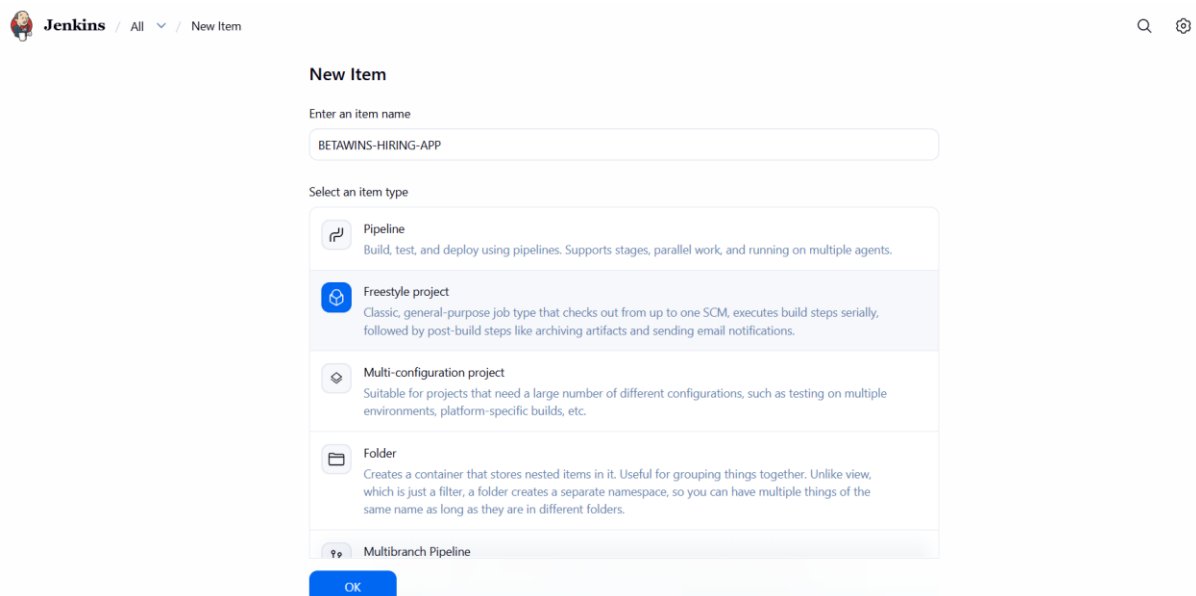
JENKINS 04

1) Setup Jenkins CI/CD pipeline using freestyle job using Docker containers using below code.

Source Code: <https://github.com/betawins/hiring-app.git>

Stages:

- Git Clone
 - SonarQube Integration
 - Maven Compilation
 - Nexus Artifactory
 - Slack Notification
 - Deploy On tomcat
-
- Opened **Jenkins** → **New Item**
 - Entered job name: **BETAWINS-HIRING-APP**
 - Selected **Freestyle project**
 - Clicked **OK** to create the job



Jenkins / All / New Item

New Item

Enter an item name

BETAWINS-HIRING-APP

Select an item type

-  Pipeline
Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.
-  Freestyle project
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.
-  Multi-configuration project
Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.
-  Folder
Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different folders.

 Multibranch Pipeline

OK

- Restricted the job to run on a specific node using **Label Expression**
- Entered label: master
- Provided GitHub repository URL: <https://github.com/betawins/hiring-app.git>

- Credentials kept as **none** (public repository)

Jenkins / BETA WINS-HIRING-APP / Configure

Configure

- General
- Source Code Management
- Triggers
- Environment
- Build Steps
- Post-build Actions

☒ Restrict where this project can be run ?

Label Expression ?

master

Label master matches 1 node. Permissions or other restrictions provided by plugins may further reduce that list.

Advanced ▾

Source Code Management

Connect and manage your code repository to automatically pull the latest code for your builds.

☐ None

☒ Git ?

Repositories ?

Repository URL ?

https://github.com/betawins/hiring-app.git

Credentials ?

- none - ▾ + Add

Advanced ▾

Save Apply

- Added **Build Step** → **Execute SonarQube Scanner**
- Selected JDK from Jenkins configuration
- Entered **Analysis properties**:
- sonar.projectKey=Test
- sonar.projectName=Test
- This enables static code analysis during the build

Jenkins / BETA WINS-HIRING-APP / Configure

Configure

- General
- Source Code Management
- Triggers
- Environment
- Build Steps
- Post-build Actions

Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment.

Execute SonarQube Scanner

JDK ?

JDK to be used for this SonarQube analysis

(Inherit From Job) ▾

Path to project properties ?

Analysis properties ?

sonar.projectKey=Test
sonar.projectName=Test

- Added **Post-build Action** → **Slack Notifications**
- Enabled notifications for:
- Build Started
- Build Success
- Build Failure

- Ensures real-time build status alerts in Slack

The screenshot shows the Jenkins configuration page for a job named 'BETAWINS-HIRING-APP'. The left sidebar contains a 'Configure' section with various tabs: General, Source Code Management, Triggers, Environment, Build Steps, and Post-build Actions (which is currently selected). The main area is titled 'Post-build Actions' and includes a description: 'Define what happens after a build completes, like sending notifications, archiving artifacts, or triggering other jobs.' Below this, there is a 'Slack Notifications' section with several checkboxes: 'Notify Build Start' (checked), 'Notify Success' (checked), 'Notify Aborted' (unchecked), 'Notify Not Built' (unchecked), 'Notify Unstable' (unchecked), 'Notify Regression' (unchecked), 'Notify Every Failure' (checked), and 'Notify Back To Normal' (unchecked). There is also an 'Advanced' dropdown menu. At the bottom, there is a '+ Add post-build action' button and 'Save' and 'Apply' buttons.

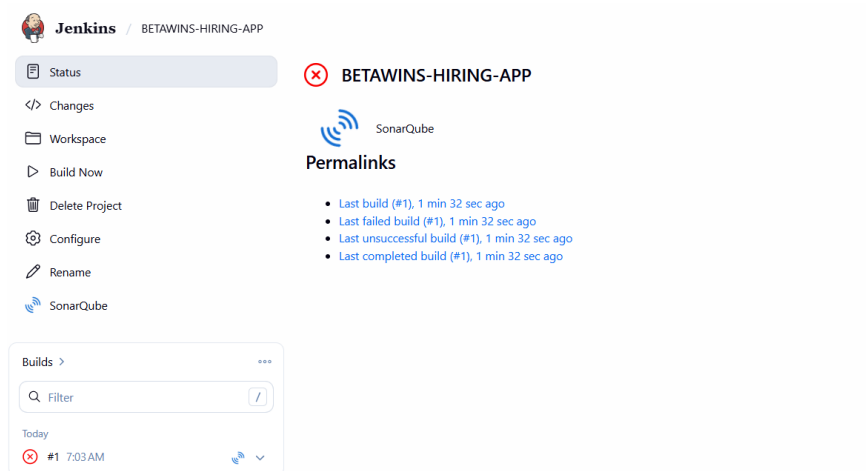
Navigated to **Manage Jenkins → System**

- Added **SonarQube Server configuration**
- Entered:
 - Name: sonarqube-server
 - Server URL: http://<sonarqube-ip>:9000
- Selected **Server authentication token**
- Saved configuration

The screenshot shows the 'Manage Jenkins' page, specifically the 'System' configuration tab. The page has a search icon and a gear icon in the top right corner. The main configuration area contains several fields: 'Name' (set to 'SonarQube'), 'Server URL' (with a default value of 'http://localhost:9000' and the entered value 'http://3.236.91.74:9000/'), and 'Server authentication token' (with a dropdown menu showing 'sonarqube-docker' and a '+ Add' button). There is also an 'Advanced' dropdown menu. At the bottom, there is a '+ Add SonarQube' button.

- Build **#1** triggered manually
- Build status: **Failed**

- Git repository and branch detected correctly
- Indicates an initial configuration or connectivity issue



- Logged into SonarQube EC2 instance
- Verified SonarQube service status:
- `./sonar.sh` status
- Confirmed SonarQube is listening on port **9000**

Validated server is running and accessible

Jenkins

BETAWINS-HIRING-APP

#1

```

> git init /var/lib/jenkins/workspace/BETAWINS-HIRING-APP # timeout=10
Fetching upstream changes from https://github.com/betawins/hiring-app.git
> git --version # timeout=10
> git --version # 'git version 2.50.1'
> git fetch --tags --force --progress -- https://github.com/betawins/hiring-app.git +refs/heads/*:refs/remotes/origin/* # timeout=10
> git config remote.origin.url https://github.com/betawins/hiring-app.git # timeout=10
> git config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeout=10
Avoid second fetch
> git rev-parse refs/remotes/origin/main^{commit} # timeout=10
Checking out Revision 9405b107fdffdf68e69fe953ea1f9669784f9d6a (refs/remotes/origin/main)
> git config core.sparsecheckout # timeout=10
> git checkout -f 9405b107fdffdf68e69fe953ea1f9669784f9d6a # timeout=10
Commit message: "Update README.md"
First time build. Skipping changelog.
[BETAWINS-HIRING-APP] $ /opt/sonar-scanner/bin/sonar-scanner -Dsonar.host.url=http://3.236.91.74:9000/ ***** -Dsonar.projectKey=Test -Dsonar.projectName=BETAWINS-HIRING-APP
Dsonar.projectBaseDir=/var/lib/jenkins/workspace/BETAWINS-HIRING-APP
07:03:32.172 INFO Scanner configuration file: /opt/sonar-scanner/conf/sonar-scanner.properties
07:03:32.180 INFO Project root configuration file: NONE
07:03:32.199 INFO SonarScanner CLI 8.0.1.6346
07:03:32.206 INFO Linux 6.18.39-79.141.amzn2023.x86_64 amd64
07:03:40.740 ERROR Failed to query server version: Call to URL [http://3.236.91.74:9000/api/v2/analysis/version] failed: HTTP connect timed out
07:03:40.741 INFO EXECUTION FAILURE
07:03:40.741 INFO Total time: 8.572s
WARN: Unable to locate 'report-task.txt' in the workspace. Did the SonarScanner succeed?
ERROR: SonarQube scanner exited with non-zero code: 1
[Slack Notifications] will send OnEveryFailureNotification because build matches and user preferences allow it
Finished: FAILURE

```

Hari

An offer awaits

Find a conversation...

Huddles

Drafts & sent

Directories

Starred

Drag and drop important stuff here

Channels

all-hari

build_notification

new-channel

social

Add channels

build_notification

Invite teammates

Huddle

Messages

Add canvas

stage_1 - #4 Started by user BOORA SRIHARI (Open)

stage_1 - #4 Success after 19 sec (Open)

Jenkins App: 7:14 PM

stage_2 - #3 Started by user BOORA SRIHARI (Open)

stage_2 - #3 Success after 12 sec (Open)

Today

Jenkins App: 12:33 PM

BETAWINS-HIRING-APP - #1 Started by user BOORA SRIHARI (Open)

BETAWINS-HIRING-APP - #1 Failure after 20 sec (Open)

Hello, team!

Oh, hello!

- Build #3 triggered after fixing issues
- Build status: **Success**
- Git checkout and SonarQube scan completed successfully

Jenkins

BETAWINS-HIRING-APP

Status

Changes

Workspace

Build Now

Delete Project

Configure

Rename

SonarQube

BETAWINS-HIRING-APP

Test

Passed

server-side processing:

Success

SonarQube Quality Gate

Last build (#3), 8 min 32 sec ago

Last stable build (#3), 8 min 32 sec ago

Last successful build (#3), 8 min 32 sec ago

Last failed build (#2), 16 min ago

Last unsuccessful build (#2), 16 min ago

Last completed build (#3), 8 min 32 sec ago

BETAWINS-HIRING-APP

Builds

Filter

Today

#3 7:28 AM

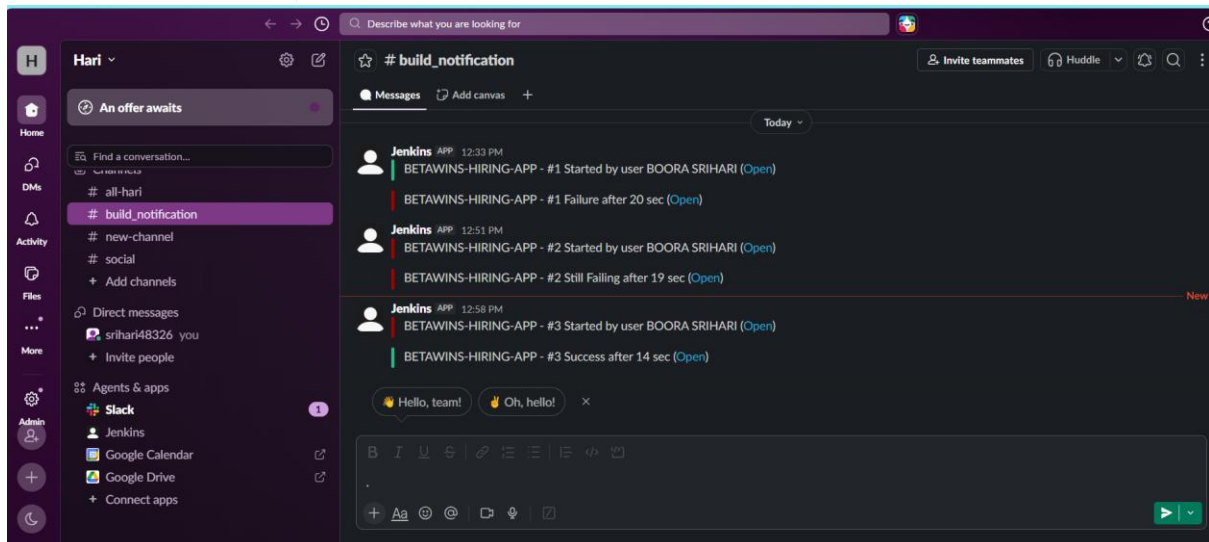
#2 7:21 AM

#1 7:03 AM

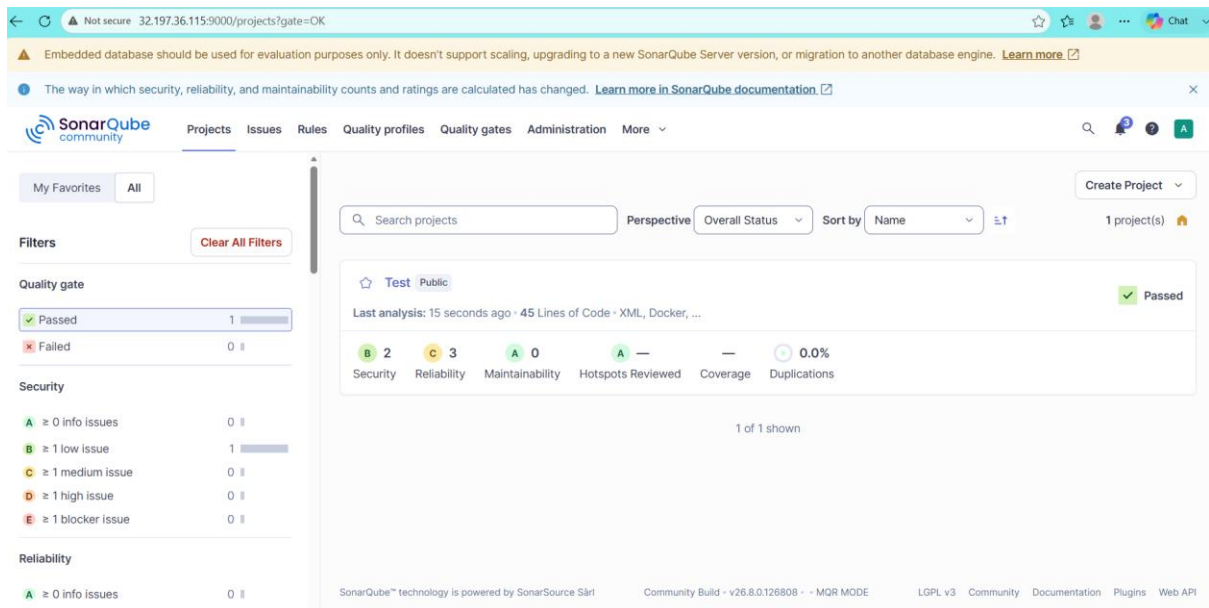
```
Jenkins / BETAWINS-HIRING-APP / #3

07:29:03.964 INFO 5/5 source files have been analyzed for the text and secrets analysis
07:29:03.967 INFO Sensor TextAndSecretsSensor [text] (done) | time=485ms
07:29:03.967 INFO ----- Run sensors on project
07:29:04.261 INFO Sensor IaC Project Sensor [iac]
07:29:04.266 INFO Sensor IaC Project Sensor [iac] (done) | time=2ms
07:29:04.266 INFO Sensor JavaProjectSensor [java]
07:29:04.266 INFO Sensor JavaProjectSensor [java] (done) | time=1ms
07:29:04.266 INFO Sensor Zero Coverage Sensor
07:29:04.266 INFO Sensor Zero Coverage Sensor (done) | time=0ms
07:29:04.266 INFO ----- Gather SCA dependencies on project
07:29:04.274 INFO Dependency analysis skipped
07:29:04.276 INFO SCM Publisher SCM provider for this project is: git
07:29:04.278 INFO SCM Publisher 4 source files to be analyzed
07:29:04.588 INFO SCM Publisher 4/4 source files have been analyzed (done) | time=309ms
07:29:04.590 INFO CPD Executor 1 file had no CPD blocks
07:29:04.591 INFO CPD Executor Calculating CPD for 0 files
07:29:04.593 INFO CPD Executor CPD calculation finished (done) | time=0ms
07:29:04.598 INFO SCM revision ID '9405b107dfdf68e69fe953ea1f9669784f9d6a'
07:29:04.809 INFO Analysis report generated in 198ms, dir size=290.3 kB
07:29:04.831 INFO Analysis report compressed in 23ms, zip size=34.9 kB
07:29:05.190 INFO Analysis report uploaded in 357ms
07:29:05.192 INFO ANALYSIS SUCCESSFUL, you can find the results at: http://32.197.36.115:9000/dashboard?id=Test
07:29:05.192 INFO Note that you will be able to access the updated dashboard once the server has processed the submitted analysis report
07:29:05.192 INFO More about the report processing at http://32.197.36.115:9000/api/ce/task?id=a102809c-79dc-4846-9c24-25fa0000ba85
07:29:05.200 INFO Analysis total time: 5.190 s
07:29:05.203 INFO SonarScanner Engine completed successfully
07:29:05.247 INFO EXECUTION SUCCESS
07:29:05.247 INFO Total time: 13.785s

[Slack Notifications] found #2 as previous completed, non-aborted build
[Slack Notifications] will send OnSuccessNotification because build matches and user preferences allow it
Finished: SUCCESS
```



- Opened SonarQube web UI
- Two projects displayed (Test, Sabear)
- Quality Gate status: Passed
- Metrics visible:
 - Bugs
 - Vulnerabilities
 - Code Smells
 - Duplications
- Confirms successful Jenkins–SonarQube integration



- Logged into the Jenkins EC2 server as **root**
- Navigated to /opt directory
- Downloaded **Apache Maven 3.9.12** using `wget https://dlcdn.apache.org/maven/maven-3/3.9.12/binaries/apache-maven-3.9.12-bin.tar.gz`
- Verified the Maven archive (apache-maven-3.9.12-bin.tar.gz) was downloaded successfully

```
[ec2-user@ip-172-31-4-173 ~]$ sudo su
[root@ip-172-31-4-173 ec2-user]# cd /opt/
[root@ip-172-31-4-173 opt]# ls
aws scripts sonar-scanner-cli-8.0.1.6346-linux-x64.zip sonar_scanner
[root@ip-172-31-4-173 opt]# wget https://dlcdn.apache.org/maven/maven-3/3.9.12/binaries/apache-maven-3.9.12-bin.tar.gz
--2026-08-18 07:55:48-- https://dlcdn.apache.org/maven/maven-3/3.9.12/binaries/apache-maven-3.9.12-bin.tar.gz
Resolving dlcdn.apache.org (dlcdn.apache.org)... 151.101.2.132, 2a04:4e42::644
Connecting to dlcdn.apache.org (dlcdn.apache.org)|151.101.2.132|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 9278065 (8.8M) [application/x-gzip]
Saving to: 'apache-maven-3.9.12-bin.tar.gz'

apache-maven-3.9.12-bin.tar.gz 100%[=====] 8.85M --.-KB/s in 0.04s

2026-08-18 07:55:48 (235 MB/s) - 'apache-maven-3.9.12-bin.tar.gz' saved [9278065/9278065]

[root@ip-172-31-4-173 opt]# tar xvf apache-maven-3.9.12-bin.tar.gz
apache-maven-3.9.12/README.txt
apache-maven-3.9.12/LICENSE
apache-maven-3.9.12/NOTICE
apache-maven-3.9.12/lib/
apache-maven-3.9.12/lib/aopalliance.license
apache-maven-3.9.12/lib/asm.license
apache-maven-3.9.12/lib/commons-cli.license
apache-maven-3.9.12/lib/commons-codec.license
apache-maven-3.9.12/lib/error_prone_annotations.license
apache-maven-3.9.12/lib/failureaccess.license
apache-maven-3.9.12/lib/gson.license
apache-maven-3.9.12/lib/guava.license
apache-maven-3.9.12/lib/guice.license
```

i-038802f91ba3b02fa (Jenkin server)
PublicIPs: 32.195.69.15 PrivateIPs: 172.31.4.173

- Extracted Maven archive:
- `tar -xvf apache-maven-3.9.12-bin.tar.gz`
- Verified extracted directory apache-maven-3.9.12
- Navigated through Maven directories (bin, lib, etc.)
- Opened .bash_profile using vi for environment configuration

```

[root@ip-172-31-4-173 ~]# cd /opt/
[root@ip-172-31-4-173 opt]# ls
apache-maven-3.9.16  apache-maven-3.9.16-bin.tar.gz  aws  scripts  sonar-scanner-cli-8.0.1.6346-linux-x64.zip  sonar_scanner
[root@ip-172-31-4-173 opt]# cd apache-maven-3.9.16
[root@ip-172-31-4-173 apache-maven-3.9.16]# ls
LICENSE NOTICE README.txt bin boot conf lib
[root@ip-172-31-4-173 apache-maven-3.9.16]# cd bin
[root@ip-172-31-4-173 bin]# ls
m2.conf mvn mvn.cmd mvnDebug mvnDebug.cmd mvnyjp
[root@ip-172-31-4-173 bin]# cd
[root@ip-172-31-4-173 ~]# vi .bash_profile
[root@ip-172-31-4-173 ~]# vi .bash_profile
[root@ip-172-31-4-173 ~]#

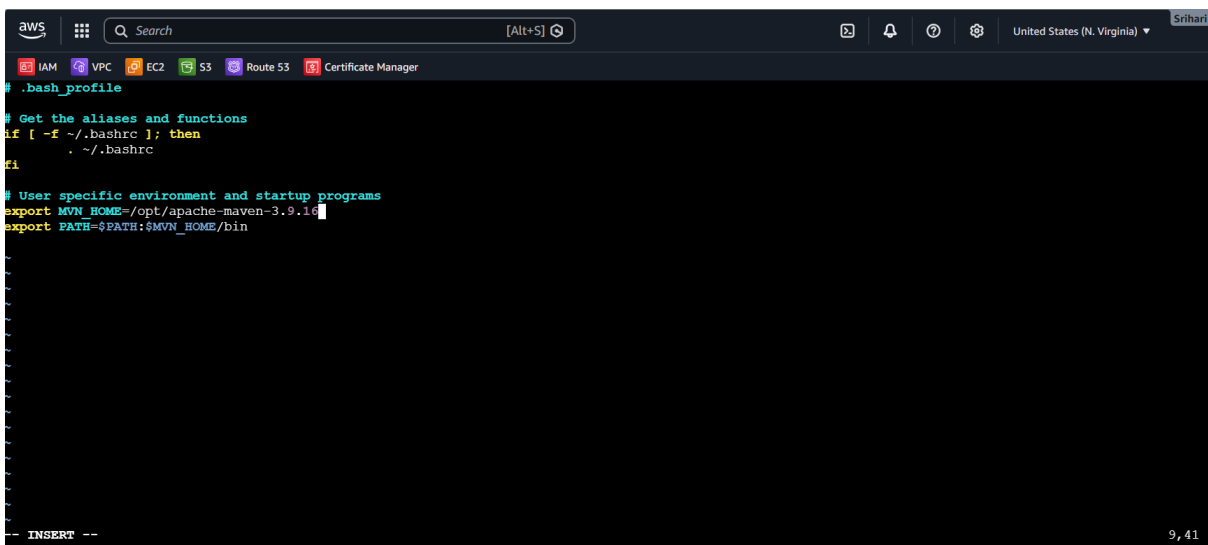
```

i-038802f91ba3b02fa (Jenkin server)

PublicIPs: 32.195.69.15 PrivateIPs: 172.31.4.173

- Edited ~/.bash_profile
- Added Maven environment variables:
 - export MVN_HOME=/opt/apache-maven-3.9.12
 - export PATH=\$PATH:\$MVN_HOME/bin

Prepared the system to recognize mvn command globally



```

aws
IAM VPC EC2 S3 Route 53 Certificate Manager
Search [Alt+S]
United States (N. Virginia) Srihari

# .bash_profile

# Get the aliases and functions
if [ -f ~/.bashrc ]; then
    . ~/.bashrc
fi

# User specific environment and startup programs
export MVN_HOME=/opt/apache-maven-3.9.16
export PATH=$PATH:$MVN_HOME/bin

-- INSERT --
9, 41

```

i-038802f91ba3b02fa (Jenkin server)

PublicIPs: 32.195.69.15 PrivateIPs: 172.31.4.173

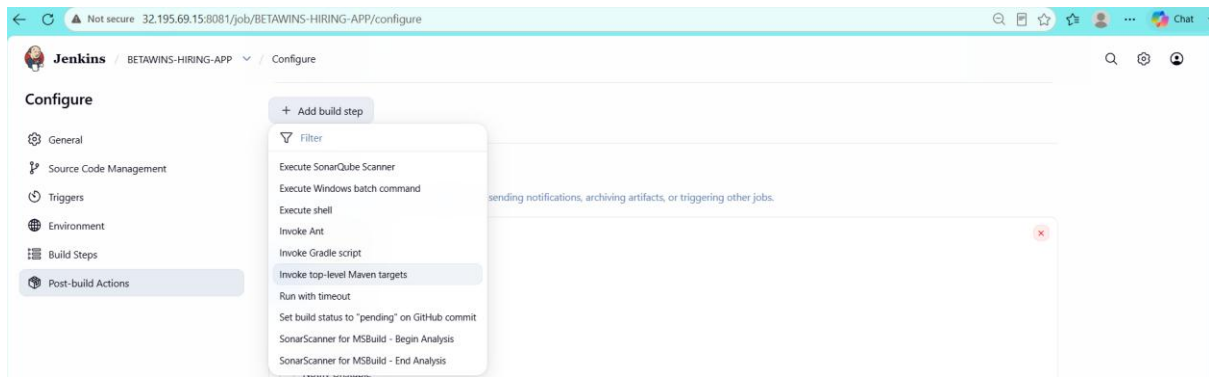
- Reloaded environment variables:
- source ~/.bash_profile
- Verified Maven installation:
 - mvn -version
 - Confirmed:
 - Apache Maven 3.9.12
 - Java 17 (Amazon Corretto)
 - Linux OS detected correctly

```
[root@ip-172-31-4-173 ~]# vi .bash_profile
[root@ip-172-31-4-173 ~]# source ~/.bash_profile
[root@ip-172-31-4-173 ~]# mvn -version
Apache Maven 3.9.16 (2bdd9fddda4b155ebf8000e807eb73fd829a51d5)
Maven home: /opt/apache-maven-3.9.16
Java version: 21.0.12, vendor: Amazon.com Inc., runtime: /usr/lib/jvm/java-21-amazon-corretto.x86_64
Default locale: en, platform encoding: UTF-8
OS name: "linux", version: "6.18.39-79.141.amzn2023.x86_64", arch: "amd64", family: "unix"
[root@ip-172-31-4-173 ~]#
```

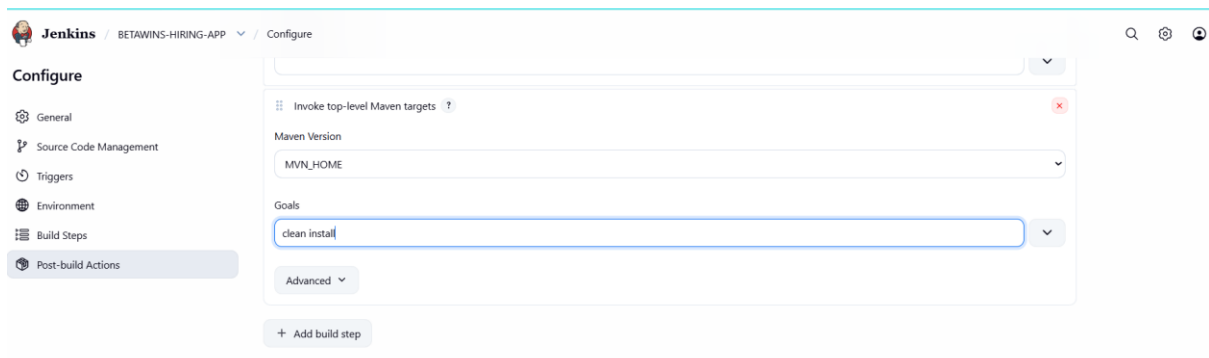
- Opened **Jenkins → Manage Jenkins → Tools**
- Navigated to **Maven installations**
- Added a new Maven installation
- Set:
 - Name: MVN_HOME
 - MAVEN_HOME path: /opt/apache-maven-3.9.12
- Disabled “Install automatically”
- Saved configuration

The screenshot shows the Jenkins web interface. At the top, there's a navigation bar with 'Jenkins', 'Manage Jenkins', and 'Tools'. Below this, there's a section for 'Maven installations'. A button '+ Add Maven' is visible. The main area shows a configuration for a Maven tool named 'Maven'. The 'Name' field is set to 'MVN_HOME' and the 'MAVEN_HOME' field is set to '/opt/apache-maven-3.9.16'. The 'Install automatically' checkbox is unchecked. At the bottom, there are 'Save' and 'Apply' buttons.

- Opened Jenkins job **BETAWINS-HIRING-APP → Configure**
- Navigated to **Build Steps**
- Clicked **Add build step**
- Selected **Invoke top-level Maven targets**



- Selected Maven Version: MVN_HOME
- Entered Maven goals:
 - clean install
- This enables:
 - Cleaning previous builds
 - Compiling code
 - Packaging the application (WAR/JAR)



- Build #5 triggered manually
- Git repository fetched successfully
- Build started without errors

The Jenkins dashboard for the project **BETAWINS-HIRING-APP** displays the following information:

- Status:** A green checkmark indicates the project is in a successful state.
- SonarQube Quality Gate:** The test result is **Passed**, with server-side processing marked as **Success**.
- Permalinks:** A list of links for various build states:
 - Last build (#5), 3 hr 6 min ago
 - Last stable build (#5), 3 hr 6 min ago
 - Last successful build (#5), 3 hr 6 min ago
 - Last failed build (#2), 4 hr 2 min ago
 - Last unsuccessful build (#2), 4 hr 2 min ago
 - Last completed build (#5), 3 hr 6 min ago
- Builds:** A table showing the last five builds:

Build Number	Status	Time
#5	Success	8:16 AM
#4	Success	7:46 AM
#3	Success	7:28 AM
#2	Failed	7:21 AM
#1	Failed	7:03 AM

- Maven dependencies downloaded from Maven Central
- Application built successfully
- WAR file generated and installed in Jenkins workspace
- Final build status:
- **BUILD SUCCESS**

The Jenkins console output for build #5 shows the following details:

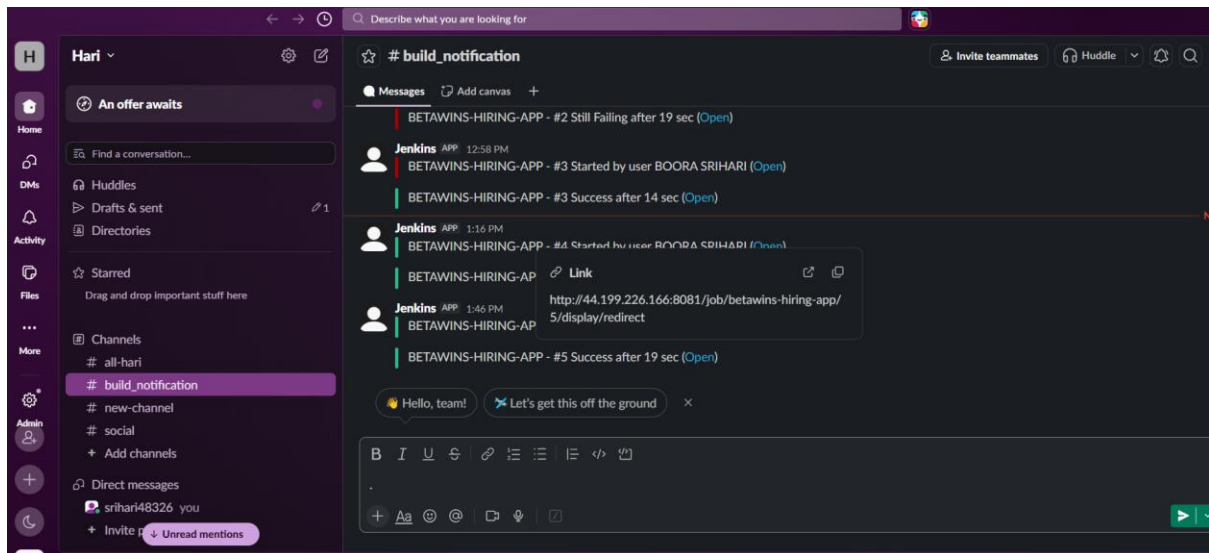
```

Progress (4): 126/193 kB | 90/157 kB
Progress (2): 126/193 kB | 106/157 kB
Progress (2): 126/193 kB | 123/157 kB
Progress (2): 126/193 kB | 139/157 kB
Progress (2): 126/193 kB | 156/157 kB
Progress (2): 126/193 kB | 157 kB
Progress (2): 142/193 kB | 157 kB

Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-api/1.9.22/maven-resolver-api-1.9.22.jar (157 kB at 9.8 MB/s)
Progress (1): 159/193 kB
Progress (1): 175/193 kB
Progress (1): 192/193 kB
Progress (1): 193 kB

Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-utils/4.0.1/plexus-utils-4.0.1.jar (193 kB at 12 MB/s)
[INFO] Installing /var/lib/jenkins/workspace/BETAWINS-HIRING-APP/pom.xml to /var/lib/jenkins/.m2/repository/in/javahome/hiring/0.1/hiring-0.1.pom
[INFO] Installing /var/lib/jenkins/workspace/BETAWINS-HIRING-APP/target/hiring.war to /var/lib/jenkins/.m2/repository/in/javahome/hiring/0.1/hiring-0.1.war
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 5.777 s
[INFO] Finished at: 2026-08-18T08:17:12Z
[INFO] -----
[Slack Notifications] found #4 as previous completed, non-aborted build
[Slack Notifications] will send OnSuccessNotification because build matches and user preferences allow it
Finished: SUCCESS
  
```

- Slack channel received Jenkins notifications:
- Build #2 – Failed
- Build #3 – Success
- Build #4 – Success
- Confirms Slack–Jenkins integration is working correctly
- Notifications triggered for both failure and success events



- Logged into the **Jenkins** server as root
- Navigated to Jenkins home directory:
- `cd /var/lib/jenkins`
- Listed Jenkins configuration and plugin files
- Verified presence of:
 - Jenkins core configuration XML files
 - Plugin configurations
 - workspace/ directory (job workspaces)

```
[root@ip-172-31-4-173 ec2-user]# cd
[root@ip-172-31-4-173 ~]# cd /var/lib/jenkins
[root@ip-172-31-4-173 jenkins]# ls
com.cloudbees.hudson.plugins.folder.config.AbstractFolderConfiguration.xml  jenkins.mvn.GlobalMavenConfig.xml
config.xml                                                                    jenkins.plugins.git.GitHooksConfiguration.xml
credentials.xml                                                                jenkins.plugins.slack.SlackNotifier.xml
fingerprints                                                                    jenkins.security.QueueItemAuthenticatorConfiguration.xml
github-plugin-configuration.xml                                                jenkins.security.ResourceDomainConfiguration.xml
hudson.model.UpdateCenter.xml                                                  jenkins.security.UpdateSiteWarningsConfiguration.xml
hudson.plugins.build_timeout.global.GlobalTimeoutConfiguration.xml            jenkins.security.apitoken.ApiTokenPropertyConfiguration.xml
hudson.plugins.build_timeout.operations.BuildStepOperation.xml                jenkins.tasks.filters.EnvVarsFilterGlobalConfiguration.xml
hudson.plugins.email_ext.ExtendedEmailPublisher.xml                           jenkins.telemetry.Correlator.xml
hudson.plugins.git.GitSCM.xml                                                  jobs
hudson.plugins.git.GitTool.xml                                                 logs
hudson.plugins.gradle.Gradle.xml                                                nodeMonitors.xml
hudson.plugins.gradle.enriched.EnrichedSummaryConfig.xml                     nodes
hudson.plugins.gradle.injection.InjectionConfig.xml                           org.jenkinsci.plugins.displayurlapi.DefaultDisplayURLProviderGlobalConfiguration.xml
hudson.plugins.sonar.MsBuildSQRunnerInstallation.xml                           org.jenkinsci.plugins.gitclient.GitHostKeyVerificationConfiguration.xml
hudson.plugins.sonar.SonarGlobalConfiguration.xml                             org.jenkinsci.plugins.gitclient.JGitApacheTool.xml
hudson.plugins.sonar.SonarRunnerInstallation.xml                               org.jenkinsci.plugins.gitclient.JGitTool.xml
```

- Navigated to Jenkins workspace:
- `cd /var/lib/jenkins/workspace/BETAWINS-HIRING-APP`
- Listed project files:
 - Jenkinsfile
 - pom.xml
 - src/
 - target/
- Confirmed Maven build artifacts exist in target/

```
[root@ip-172-31-4-173 jenkins]# cd workspace/
[root@ip-172-31-4-173 workspace]# ls
BETAWINS-HIRING-APP  first-job-01  jenkins  jenkins@tmp  sample-1  sample-2  sample-3  stage_1  stage_2  webhook  webhook@tmp
[root@ip-172-31-4-173 workspace]# cd BETAWINS-HIRING-APP
[root@ip-172-31-4-173 BETAWINS-HIRING-APP]# ls
Dockerfile  Jenkinsfile  README.md  'Untitled Diagram.drawio'  jenkinsfile-cicd  pom.xml  src  target
[root@ip-172-31-4-173 BETAWINS-HIRING-APP]# cd target
[root@ip-172-31-4-173 target]# ls
hiring  hiring.war  maven-archiver
[root@ip-172-31-4-173 target]#
```

i-038802f91ba3b02fa (Jenkin server)

PublicIPs: 32.195.69.15 PrivateIPs: 172.31.4.173

- Navigated to Maven local repository:
- `cd /var/lib/jenkins/.m2/repository`
- Listed downloaded dependencies:
- `org`
- `junit`
- `commons-io`
- `javax`
- Confirms Maven dependencies were downloaded successfully during build

```
[root@ip-172-31-4-173 target]# cd
[root@ip-172-31-4-173 ~]# cd /var/lib/jenkins/.m2/
[root@ip-172-31-4-173 .m2]# ls
repository
[root@ip-172-31-4-173 .m2]# cd repository/
[root@ip-172-31-4-173 repository]# ls
com  commons-io  in  javax  junit  org
[root@ip-172-31-4-173 repository]# ll
total 0
drwxr-xr-x. 4 jenkins jenkins 40 Aug 18 08:17 com
drwxr-xr-x. 3 jenkins jenkins 24 Aug 18 08:17 commons-io
drwxr-xr-x. 3 jenkins jenkins 22 Aug 18 08:17 in
drwxr-xr-x. 3 jenkins jenkins 20 Aug 18 08:17 javax
drwxr-xr-x. 3 jenkins jenkins 19 Aug 18 08:17 junit
drwxr-xr-x. 10 jenkins jenkins 116 Aug 18 08:17 org
[root@ip-172-31-4-173 repository]#
```

i-038802f91ba3b02fa (Jenkin server)

PublicIPs: 32.195.69.15 PrivateIPs: 172.31.4.173

- Opened **Amazon EC2** → **Launch Instance**
- Set instance name: **NEXUS-SERVER**
- Selected **Amazon Linux 2023 AMI**
- Chose instance type
- Configured:
- Security Group
- Storage
- Clicked **Launch Instance**

EC2

Instances

Launch an instance

Launch an instance

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags

Name

NEXUS-SERVER

Add additional tags

Application and OS Images (Amazon Machine Image)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Q Search our full catalog including 1000s of application and OS images

Recents

Quick Start

Amazon Linux

macOS

Ubuntu

Windows

Red Hat

SUSE Linux

Debian

Browse more AMIs

Summary

Number of instances

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.12...[read more](#)

Virtual server type (instance type)

c7i-flex.large

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Cancel

Launch instance

Preview code

```

[root@ip-172-31-7-120 ec2-user]# yum install java-21-amazon-corretto -y
No such command: install. Please use /usr/bin/yum --help
it could be a YUM plugin command, try: "yum install 'dnf-command(install)'"
[root@ip-172-31-7-120 ec2-user]# sudo dnf install java-21-amazon-corretto-devel -y
Amazon Linux 2023 Kernel Livepatch repository
Dependencies resolved.
694 kB/s | 69 kB | 00:00

```

Package	Architecture	Version	Repository	Size
Installing:				
java-21-amazon-corretto-devel	x86_64	1:21.0.12+8-1.amzn2023.1	amazonlinux	149 k
Installing dependencies:				
cairo	x86_64	1.18.0-4.amzn2023.0.3	amazonlinux	717 k
fontsfilesystem	noarch	1:2.0.5-12.amzn2023.0.2	amazonlinux	9.5 k
google-noto-fonts-common	noarch	20240401-1.amzn2023.0.2	amazonlinux	17 k

Step 2: Install Docker

For Amazon Linux:

```
sudo dnf install docker -y
```

Start Docker:

```
sudo systemctl start docker
```

Enable it:

```
sudo systemctl enable docker
```

Check:

```
sudo systemctl status docker
```

Step 3: Allow your user to use Docker

```
sudo usermod -aG docker ec2-user
```

Then logout and login again.

Check:

```
docker ps
```

Part 2 — Run Nexus using Docker

I recommend using a Docker volume because Nexus stores its configuration, logs and repository data under /nexus-data; Sonatype recommends persistent Docker storage for this.

Create volume:

docker volume create nexus-data

Run Nexus:

```
docker run -d \
  --name nexus \
  -p 8081:8081 \
  -v nexus-data:/nexus-data \
  sonatype/nexus3:latest
```

Check:

docker ps

You should see:

nexus sonatype/nexus3 Up ...

```
[ec2-user@ip-172-31-7-120 ~]$ sudo su
[root@ip-172-31-7-120 ec2-user]# sudo systemctl status docker
● docker.service - Docker Application Container Engine
   Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: disabled)
   Active: active (running) since Tue 2026-08-18 11:36:47 UTC; 56min ago
   TriggeredBy: ● docker.socket
     Docs: https://docs.docker.com
    Main PID: 33814 (dockerd)
      Tasks: 25
     Memory: 398.8M
        CPU: 7.513s
    CGroup: /system.slice/docker.service
            └─33814 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock --default-ulimit nfile=32768:65536
              └─34269 /usr/bin/docker-proxy -proto tcp -host-ip 0.0.0.0 -host-port 8081 -container-ip 172.17.0.2 -container-port 8081
                └─34275 /usr/bin/docker-proxy -proto tcp -host-ip :: -host-port 8081 -container-ip 172.17.0.2 -container-port 8081

Aug 18 11:36:47 ip-172-31-7-120.ec2.internal systemd[1]: Starting docker.service - Docker Application Container Engine...
Aug 18 11:36:47 ip-172-31-7-120.ec2.internal dockerd[33814]: time="2026-08-18T11:36:47.043326926Z" level=info msg="Starting up"
Aug 18 11:36:47 ip-172-31-7-120.ec2.internal dockerd[33814]: time="2026-08-18T11:36:47.074774030Z" level=info msg="Loading containers: start."
Aug 18 11:36:47 ip-172-31-7-120.ec2.internal dockerd[33814]: time="2026-08-18T11:36:47.354667636Z" level=info msg="Loading containers: done."
Aug 18 11:36:47 ip-172-31-7-120.ec2.internal dockerd[33814]: time="2026-08-18T11:36:47.372493325Z" level=info msg="Docker daemon" commit=6fdf0a6 containerd
Aug 18 11:36:47 ip-172-31-7-120.ec2.internal dockerd[33814]: time="2026-08-18T11:36:47.372642208Z" level=info msg="Daemon has completed initialization"
Aug 18 11:36:47 ip-172-31-7-120.ec2.internal dockerd[33814]: time="2026-08-18T11:36:47.386527518Z" level=info msg="API listen on /run/docker.sock"
Aug 18 11:36:47 ip-172-31-7-120.ec2.internal systemd[1]: Started docker.service - Docker Application Container Engine.
Aug 18 11:40:48 ip-172-31-7-120.ec2.internal dockerd[33814]: 2026/08/18 11:40:48 http: superfluous response.WriteHeader call from go.opentelemetry.io/cont

[root@ip-172-31-7-120 ec2-user]# docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
92aac6b0f1bf   sonatype/nexus3:latest             "/opt/sonatype/nexus..." 53 minutes ago Up 53 minutes 0.0.0.0:8081->8081/tcp, :::8081->8081/tcp
nexus

i-0a8056acfe3f4019f (NEXUS-SERVER)
PublicIPs: 100.24.106.233 PrivateIPs: 172.31.7.120
```

Part 3 — Open Nexus in browser

In the Nexus EC2 Security Group, add:

Type: Custom TCP

Port: 8081

Source: Your IP

Then open:

http://<NEXUS-EC2-PUBLIC-IP>:8081

You should get the Nexus Repository webpage.

Part 4 — Get the admin password

Run:

```
docker exec nexus cat /nexus-data/admin.password
```

Copy the password.

Login with:

Username: admin

Password: <password from above>

Nexus's initial admin password is stored inside the persistent data location.

```
[root@ip-172-31-7-120 ec2-user]# docker exec nexus cat /nexus-data/admin.password
b2e42a98-9a09-47b1-8c97-dcf24e20ce04[root@ip-172-31-7-120 ec2-user]#
```

i-0a8056acfe3f4019f (NEXUS-SERVER)
PublicIPs: 100.24.106.233 PrivateIPs: 172.31.7.120

Part 5 — Create Maven repository

In Nexus:

Repositories → Create repository

Select:

maven2 (hosted)

Configure:

Name:

maven-releases

For your first assignment, you can use:

Version policy:

Release

Deployment policy:

Allow redeploy

Click:

Create repository

Your repository URL will be:

http://<NEXUS-EC2-IP>:8081/repository/maven-releases/

sonatype nexus repository
Community Edition

Search components or CVEs...

Dashboard

Search

Browse

Upload

Malware Risk

Settings

Usage Center

Usage below limits

Monitor this instance's usage to ensure your deployment is appropriate for your needs. Learn more about the usage center.

Usage Metrics Overview

Total Components

0 Current40,000 Usage Limit

0 Highest Recorded Count (30 days)

Requests Per Day

0 Last 24 hours100,000 Usage Limit

0 Highest Recorded Count (30 days)

Requests Per Month

0 Total requests in August

0 Average requests (12 months)

0 Highest recorded count (12 months)

System Health Check
View system status checks

Cleanup Policies
Review component removal policies

Browse
Browse my repositories

Notification Center

You have 1 notification to review

sonatype nexus repository
Community Edition

Search components or CVEs...

Dashboard

Search

Browse

Upload

Malware Risk

Settings

Settings

Repository

Repositories

Blob Stores

Data Store

Proprietary Repositories

Content Selectors

Cleanup Policies

Routing Rules

Security

Repositories

Create repository

Filter

Name	Type	Format	Blob Store	Status	URL	Health check	Firewall Re...
maven-ce...	proxy	maven2	default	Online - R...	copy	0 0	
maven-pu...	group	maven2	default	Online	copy		
maven-rel...	hosted	maven2	default	Online	copy		
maven-sna...	hosted	maven2	default	Online	copy		
nuget-group	group	nuget	default	Online	copy		
nuget-host...	hosted	nuget	default	Online	copy		
nuget.org...	proxy	nuget	default	Online - R...	copy	0 0	

sonatype nexus repository
Community Edition

Search components or CVEs...

Dashboard

Search

Browse

Upload

Malware Risk

Settings

Settings

Repository

Repositories

Blob Stores

Data Store

Proprietary Repositories

Content Selectors

Cleanup Policies

Routing Rules

Security

Support

Repositories

Select Recipe

Create Repository: maven2 (hosted)

Name:
A unique identifier for this repository
maven-releases

Online:
☒ If checked, the repository accepts incoming requests

Maven 2

Version policy:
What type of artifacts does this repository store?
Release

Layout policy:
Validate that all paths are maven artifact or metadata paths
Strict

Content Disposition:
Add Content-Disposition header as 'Attachment' to disable some content from being inline in a browser.
Attachment

sonatype nexus repository
Community Edition

Search components or CVEs...

Dashboard

Search

Browse

Upload

Malware Risk

Settings

Settings

Repository

Repositories

Blob Stores

Data Store

Proprietary Repositories

Content Selectors

Cleanup Policies

Routing Rules

Security

Repositories

Select Recipe

Create Repository: maven2 (hosted)

Storage

Blob store:
Blob store used to store repository contents
default

Strict Content Type Validation:
☒ Validate that all content uploaded to this repository is of a MIME type appropriate for the repository format

Hosted

Deployment policy:
Controls if deployments of and updates to artifacts are allowed
Allow redeploy

Proprietary Components:
☐ Components in this repository count as proprietary for namespace conflict attacks (requires Sonatype Nexus Firewall)

The screenshot shows the Sonatype Nexus Repository Community Edition interface. On the left, the 'Settings' sidebar is open, with 'Repository' selected. The main area displays a table of repositories:

Name ↑	Type	Format	Blob Store	Status	URL	Health check	Firewall Re...
maven-central	proxy	maven2	default	Online - Ready to...		0 0	
maven-public	group	maven2	default	Online			
maven-releases	hosted	maven2	default	Online			
maven-releases-01	hosted	maven2	default	Online			
maven-snapshots	hosted	maven2	default	Online			

Below the table, a terminal window shows the output of a curl command:

```
[root@ip-172-31-4-173 ec2-user]# curl http://100.24.106.233:8081
<!DOCTYPE html>
<html lang="en" class="nx-html">
<head>
<title>Sonatype Nexus Repository</title>
<meta http-equiv="Content-Type" content="text/html; charset=utf-8"/>
<meta name="description" content="Sonatype Nexus Repository"/>
<meta http-equiv="X-UA-Compatible" content="IE=edge"/>

<link rel="icon" type="image/png" href="/static/rapture/resources/safari-favicon-32x32.png?_v=3.95.1-01&_e=COMMUNITY&_c=2026-08-11-2008-3364" size=
<link rel="mask-icon" href="/static/rapture/resources/favicon-white.svg?_v=3.95.1-01&_e=COMMUNITY&_c=2026-08-11-2008-3364" color="#00bb6c">
<link rel="icon" type="image/png" href="/static/rapture/resources/favicon.svg?_v=3.95.1-01&_e=COMMUNITY&_c=2026-08-11-2008-3364" sizes="16x16">
```

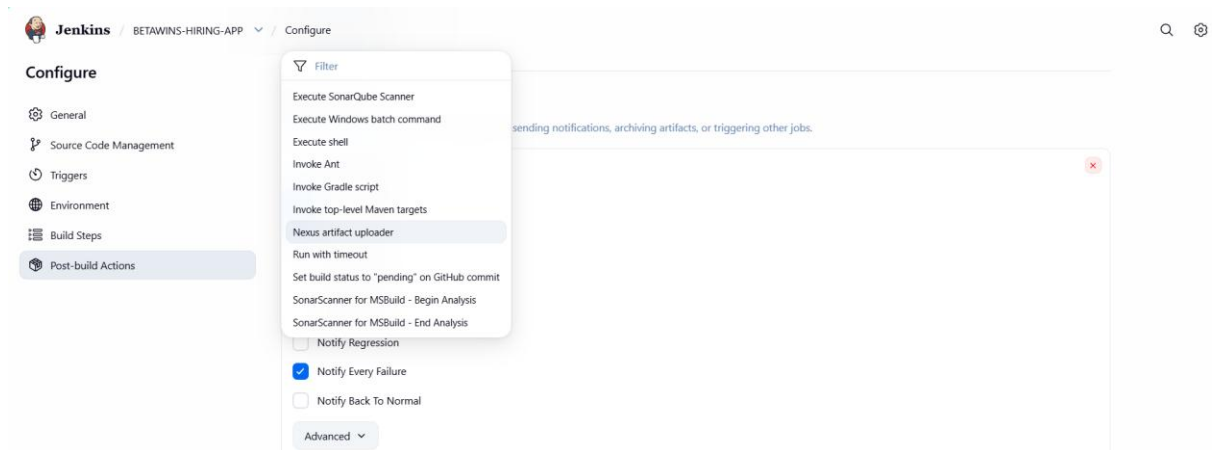
At the bottom, the server information is displayed:

i-038802f91ba3b02fa (Jenkin server)
 PublicIPs: 32.195.69.15 PrivateIPs: 172.31.4.173

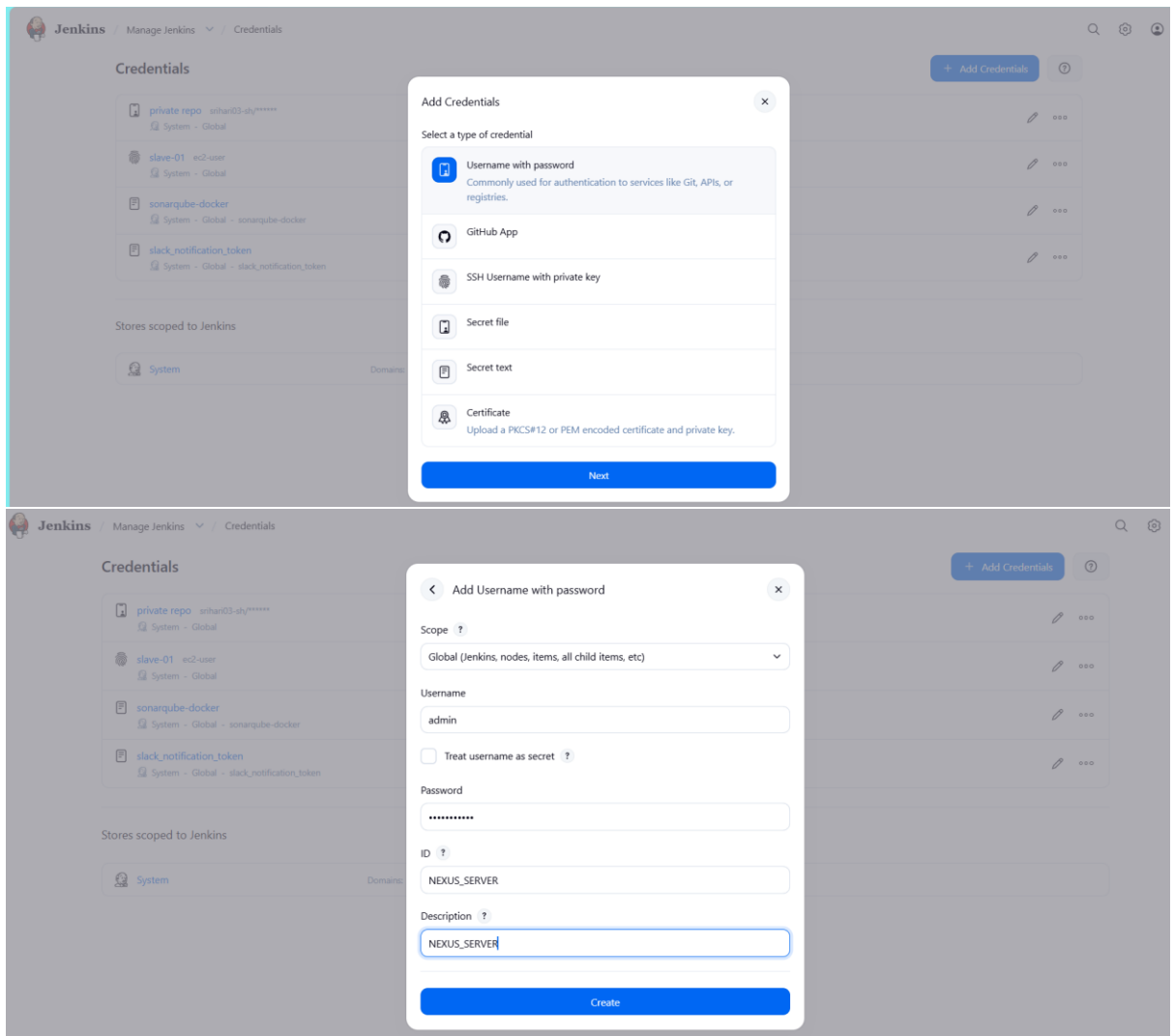
- Opened **Jenkins** → **Manage Jenkins** → **Plugins**
- Searched for **nexus**
- Selected **Nexus Artifact Uploader** plugin
- Installed the plugin successfully

The screenshot shows the Jenkins 'Manage Jenkins' → 'Plugins' page. The search bar contains 'nexus'. The 'Available plugins' list shows the 'Nexus Artifact Uploader' plugin (version 2.16) as installed. The plugin details show it is used to upload artifacts to Nexus Repository. A message indicates the plugin is up for adoption and looking for new maintainers. The release date is '3 mo 16 days ago' and the health status is '77'.

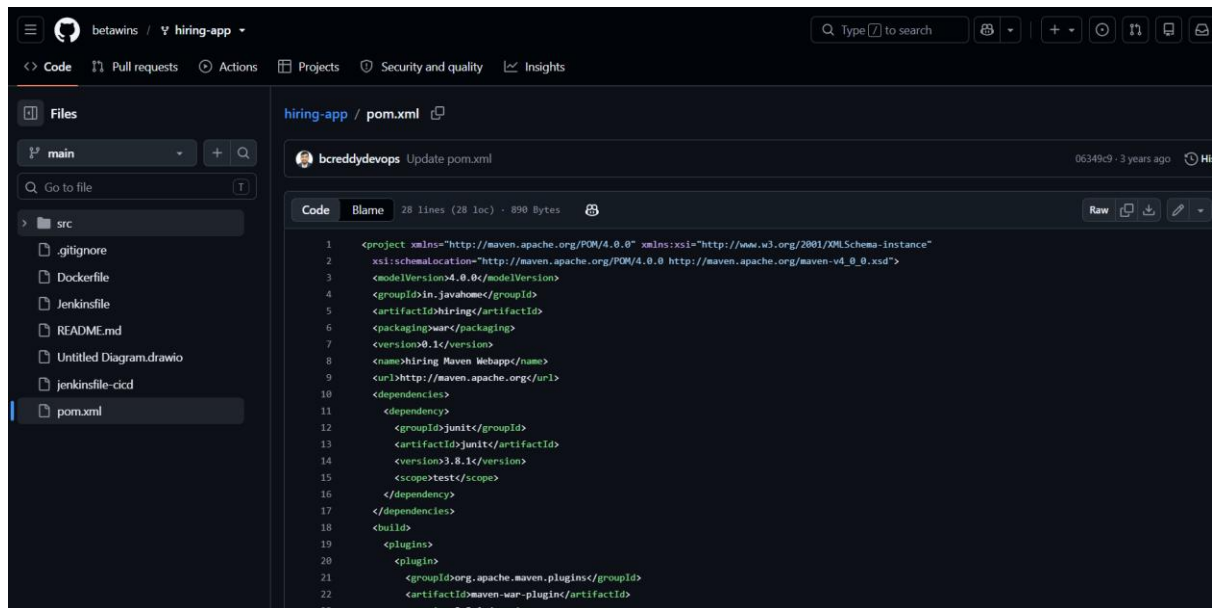
- Opened Jenkins job **BETAWINS-HIRING-APP** → **Configure**
- Navigated to **Build Steps**
- Clicked **Add build step**
- Selected **Nexus artifact uploader**



- Clicked **Add** → **Jenkins Credentials**
- Selected:
- Kind: Username with password
- Scope: Global
- Entered:
- Username: admin
- Password: Nexus admin password
- ID: NEXUS_SERVER
- Description: NEXUS_SERVER
- Saved credentials for Nexus authentication



- Opened the project **pom.xml** in **GitHub**
- Identified Maven coordinates:
 - groupId
 - artifactId
 - version
- In **Jenkins** → **Nexus artifact uploader**, entered:
 - **GroupId**: in.javahome
 - **Version**: 0.1
- Error shown: **Repository must not be empty** (repository not yet created/selected)



- Logged into **Nexus Repository Manager**
- Navigated to **Settings → Repository → Repositories**
- Viewed existing repositories:
 - maven-central (proxy)
 - maven-releases (hosted)
 - maven-snapshots (hosted)
- Confirmed no project-specific hosted repo exists yet
- Clicked **Create repository**
- Selected **maven2 (hosted)**
- This type is required to **upload artifacts from Jenkins**
- Entered repository details:
 - **Name:** hiring-app
 - **Version policy:** Release
 - **Layout policy:** Permissive
 - **Blob store:** default
 - **Deployment policy:** Allow redeploy
- Saved the repository
- Nexus hosted repository hiring-app created successfully

The screenshot shows the Sonatype Nexus Repository Community Edition web interface. The top navigation bar includes the Sonatype logo, a search bar, and user icons. The left sidebar contains a 'Settings' menu with options like Repository, Repositories, Blob Stores, Data Store, Proprietary Repositories, Content Selectors, Cleanup Policies, Routing Rules, Security, Support, and System. The main content area is titled 'Repositories / maven-releases-01' and features buttons for 'Delete repository' and 'Rebuild index'. Below these is a 'Settings' tab with the following configuration details:

- Name:** maven-releases-01
- Format:** maven2
- Type:** hosted
- URL:** <http://100.24.106.233:8081/repository/maven-releases-01/>
- Online:** ☒ If checked, the repository accepts incoming requests

Under the 'Maven 2' section, the following policies are configured:

- Version policy:** Release
- Layout policy:** Strict (selected from a dropdown menu)
- Content Disposition:** Attachment (selected from a dropdown menu)

- Returned to **Jenkins** → **Job** → **Configure** → **Nexus artifact uploader**
- Entered:
 - **Repository:** hiring-app
- Verified repository URL in Nexus UI
- Jenkins and Nexus repository names now match

Configure

- General
- Source Code Management
- Triggers
- Environment
- Build Steps**
- Post-build Actions

Nexus Details

Nexus Version

NEXUS2

Protocol

HTTP

Nexus URL ?

100.24.106.233:8081/repository/maven-releases-01/

Credentials

admin***** (NEXUS_SERVER)

+ Add

GroupId

in.javahome

Version

0.1

Repository ?

maven-releases-01

Save

Apply

Configure

- General
- Source Code Management
- Triggers
- Environment
- Build Steps
- Post-build Actions**

0.1

Repository ?

maven-releases-01

Artifacts

Artifact

X

ArtifactId

hiring

Type ?

war

Classifier ?

File ?

/var/lib/jenkins/workspace/BETAWINS-HIRING-APP/target/hiring.war

Save

Apply

Jenkins / BETAWINS-HIRING-APP / #10

Status

</> Changes

Console Output

Delete build '#10'

Git Build Data

Edit build information

Previous Build

#10 (Aug 18, 2026, 1:36:56 PM)

Started by user BOORA SRIHARI

Revision: 9405b107dfdd68e69fe953ea19669784f9d6a
Repository: <https://github.com/betawins/hiring-app.git>

- refs/remotes/origin/main

No changes.

Add description

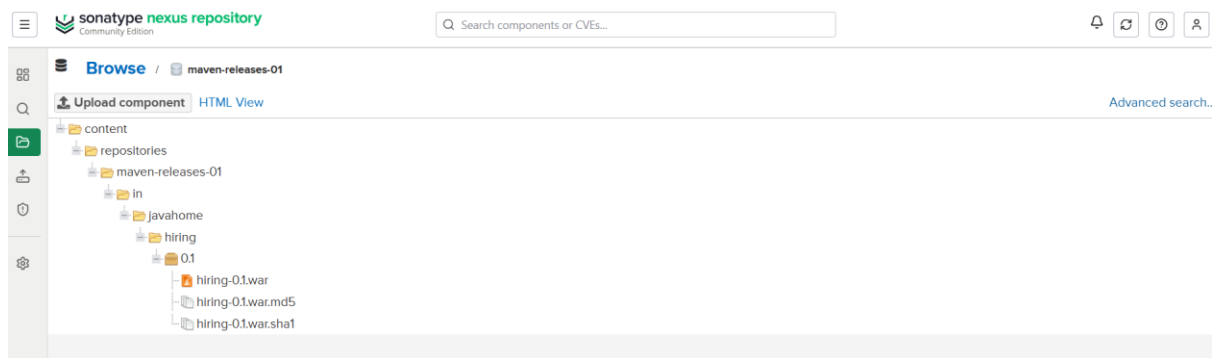
Keep this build forever

Started 3 min 26 sec ago
Took 14 sec on built-in

Jenkins / BETAWINS-HIRING-APP / #10

```
[INFO] Copying webapp resources [/var/lib/jenkins/workspace/BETAWINS-HIRING-APP/src/main/webapp]
[INFO] Building war: /var/lib/jenkins/workspace/BETAWINS-HIRING-APP/target/hiring.war
[INFO]
[INFO] --- install:3.1.4:install (default-install) @ hiring ---
[INFO] Installing /var/lib/jenkins/workspace/BETAWINS-HIRING-APP/pom.xml to /var/lib/jenkins/.m2/repository/in/javahome/hiring/0.1/hiring-0.1.pom
[INFO] Installing /var/lib/jenkins/workspace/BETAWINS-HIRING-APP/target/hiring.war to /var/lib/jenkins/.m2/repository/in/javahome/hiring/0.1/hiring-0.1.war
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 1.353 s
[INFO] Finished at: 2026-08-18T13:37:11Z
[INFO] -----
Uploading artifact hiring.war started....
GroupId: in.javahome
ArtifactId: hiring
Classifier:
Type: war
Version: 0.1
File: hiring.war
Repository:maven-releases-01
Uploading: http://100.24.106.233:8081/repository/maven-releases-01/content/repositories/maven-releases-01/in/javahome/hiring/0.1/hiring-0.1.war
100 % completed (1.7 kB / 1.7 kB).
Uploaded: http://100.24.106.233:8081/repository/maven-releases-01/content/repositories/maven-releases-01/in/javahome/hiring/0.1/hiring-0.1.war (1.7 kB at 11 kB/s)
Uploading artifact hiring.war completed.
[Slack Notifications] found #9 as previous completed, non-aborted build
[Slack Notifications] will send OnSuccessNotification because build matches and user preferences allow it
Finished: SUCCESS
```

- Logged into **Sonatype Nexus Repository Manager**
- Navigated to **Browse**
- Opened repository: **hiring-app**
- Navigated through Maven coordinates:
 - in/
 - └─ javahome/
 - └─ hiring/
 - └─ 0.1/
- Verified uploaded artifacts:
 - hiring-0.1.war
 - hiring-0.1.war.md5
 - hiring-0.1.war.sha1
- Confirms artifact upload to Nexus was successful

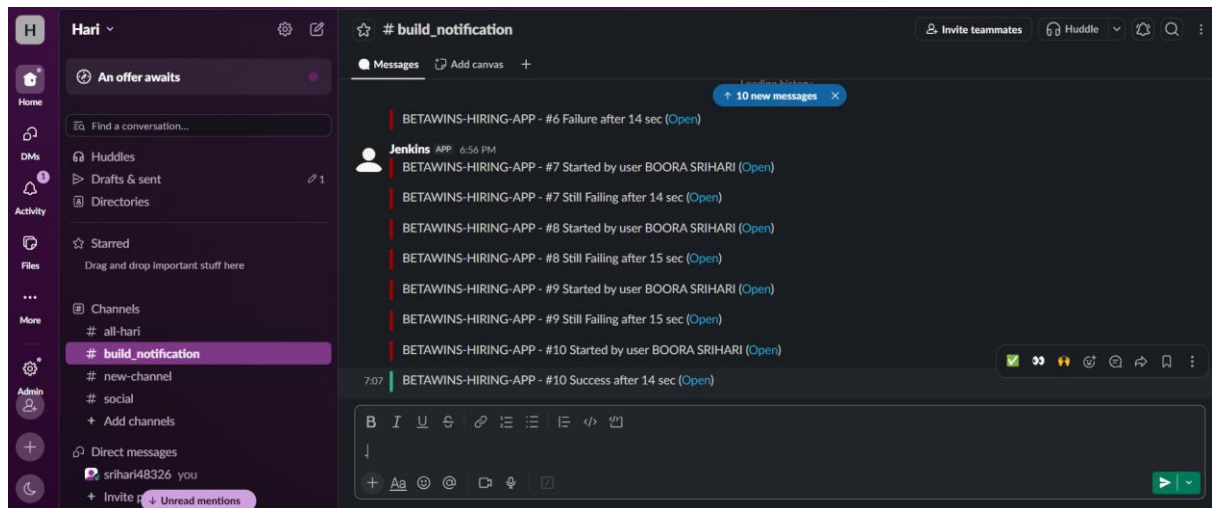


Slack channel received Jenkins notifications:

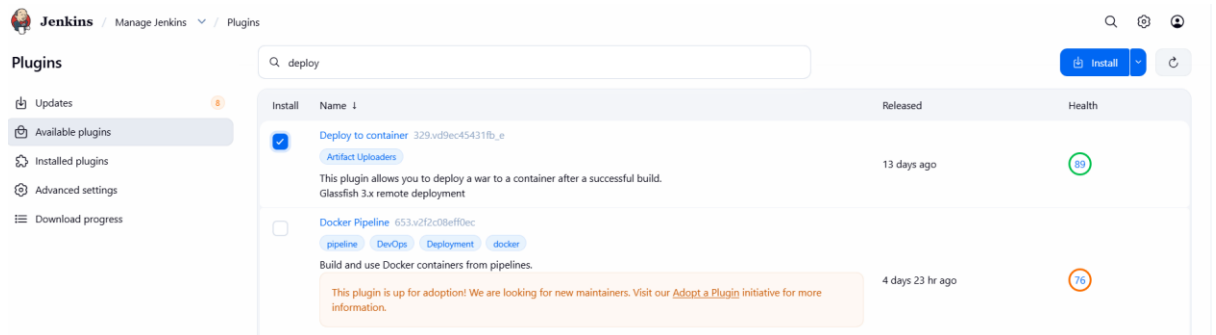
- Build #4 – Failed
- Build #5 – Success

Confirms:

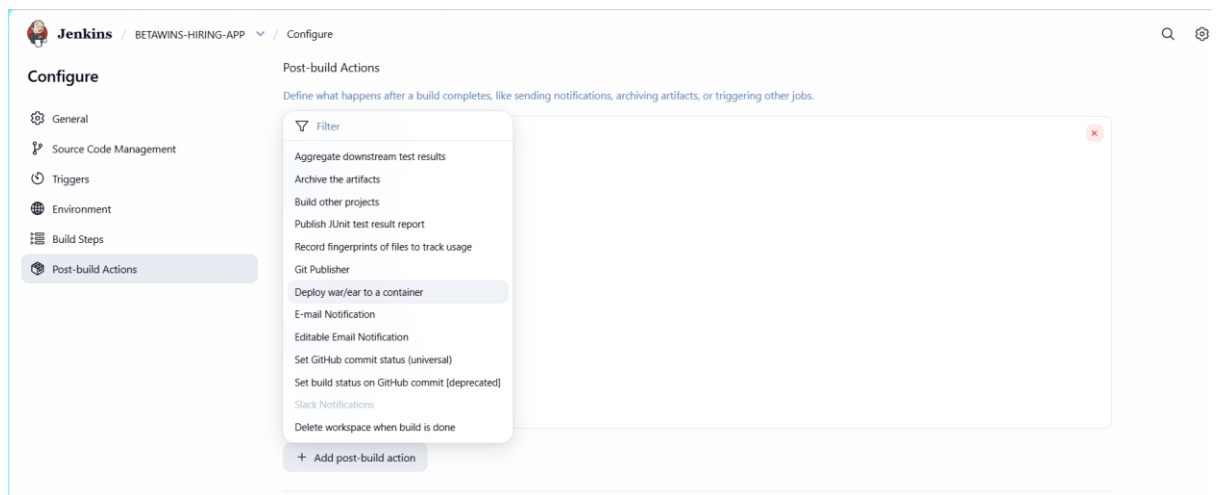
- Jenkins–Slack integration is working



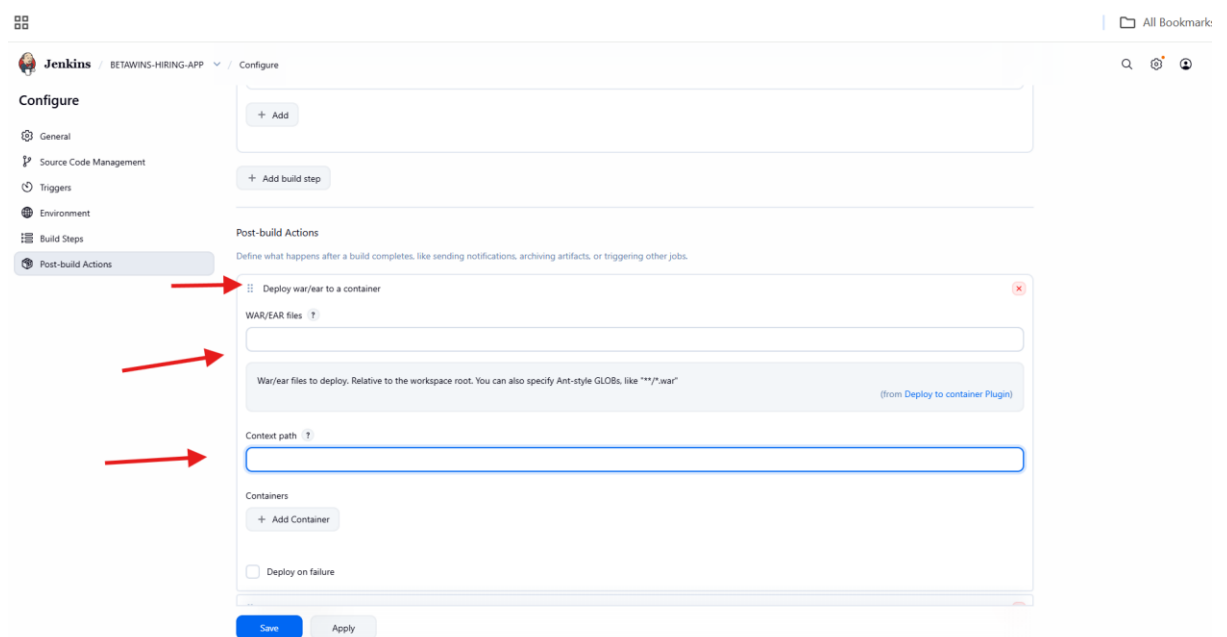
- Opened **Jenkins**
- Navigated to **Manage Jenkins** → **Plugins** → **Available plugins**
- Searched for **deploy**
- Selected **Deploy to container** plugin
- Clicked **Install**
- Plugin installed successfully for WAR deployment



- Opened Jenkins job **BETAWINS-HIRING-APP**
- Clicked **Configure**
- Navigated to **Post-build Actions**
- Clicked **Add post-build action**
- Selected **Deploy war/ear to a container**



- Opened **Deploy war/ear to a container** section
- Configuration pending (Tomcat not yet set up)



- Opened **Amazon EC2** → **Launch instance**
- Set instance name: **tomcat**
- Clicked **Launch instance**
- Logged into **tomcat EC2 instance**
- Installed Java 17 (Amazon Corretto):
- `yum install java-17-amazon-corretto-y`
- Verified Java packages installed
- Java required for Tomcat runtime
- Navigated to `/opt`
- Downloaded Tomcat archive:
- `wget https://dlcdn.apache.org/tomcat/tomcat-9/v9.0.113/bin/apache-tomcat-9.0.113.tar.gz`
- Extracted the archive:
- `tar -xvf apache-tomcat-9.0.113.tar.gz`
- Tomcat directory created successfully

```
[root@ip-172-31-7-120 ec2-user]# cd /opt
[root@ip-172-31-7-120 opt]# wget https://dlcdn.apache.org/tomcat/tomcat-9/v9.0.121/bin/apache-tomcat-9.0.121.tar.gz
--2026-08-18 14:02:49-- https://dlcdn.apache.org/tomcat/tomcat-9/v9.0.121/bin/apache-tomcat-9.0.121.tar.gz
Resolving dlcdn.apache.org (dlcdn.apache.org)... 151.101.2.132, 2a04:4e42::644
Connecting to dlcdn.apache.org (dlcdn.apache.org)[151.101.2.132]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 13178703 (13M) [application/x-gzip]
Saving to: 'apache-tomcat-9.0.121.tar.gz'

apache-tomcat-9.0.121.tar.gz      100%[=====>]  12.57M  --.-KB/s

2026-08-18 14:02:49 (302 MB/s) - 'apache-tomcat-9.0.121.tar.gz' saved [13178703/13178703]

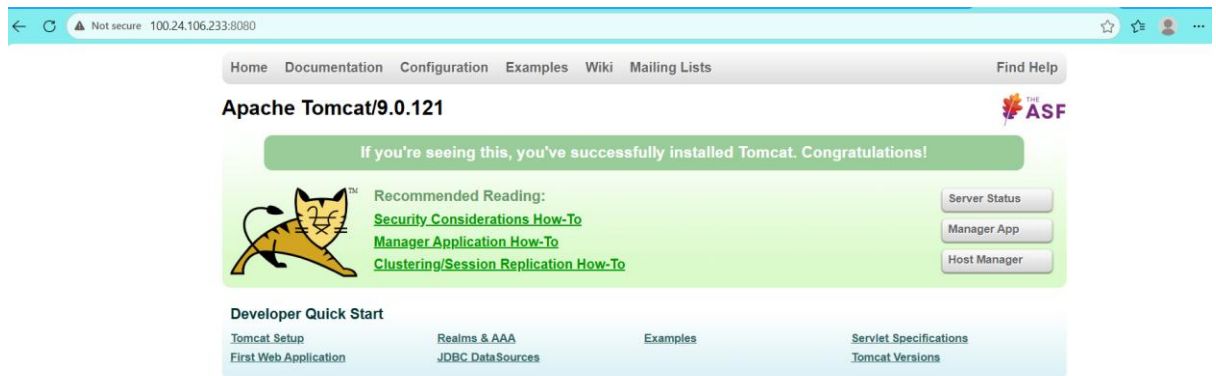
[root@ip-172-31-7-120 opt]# tar -xvzf /opt/apache-tomcat-9.0.121.tar.gz
apache-tomcat-9.0.121/conf/
apache-tomcat-9.0.121/conf/catalina.policy
apache-tomcat-9.0.121/conf/catalina.properties
apache-tomcat-9.0.121/conf/context.xml
apache-tomcat-9.0.121/conf/jaspic-providers.xml
apache-tomcat-9.0.121/conf/jaspic-providers.xsd
apache-tomcat-9.0.121/conf/logging.properties
apache-tomcat-9.0.121/conf/server.xml
apache-tomcat-9.0.121/conf/tomcat-users.xml
apache-tomcat-9.0.121/conf/tomcat-users.xsd
apache-tomcat-9.0.121/conf/web.xml
apache-tomcat-9.0.121/bin/
apache-tomcat-9.0.121/lib/
apache-tomcat-9.0.121/logs/
apache-tomcat-9.0.121/temp/

i-0a8056acfe3f4019f (NEXUS-SERVER)
PublicIPs: 100.24.106.233  PrivateIPs: 172.31.7.120
```

- Navigated to Tomcat bin directory:
- `cd /opt/apache-tomcat-9.0.113/bin`
- Started Tomcat:
- `./startup.sh`
- Tomcat started without errors

```
[root@ip-172-31-7-120 opt]# chmod +x /opt/apache-tomcat-9.0.121/bin/startup.sh shutdown.sh
chmod: cannot access 'shutdown.sh': No such file or directory
[root@ip-172-31-7-120 opt]# cd /opt/apache-tomcat-9.0.121/bin
[root@ip-172-31-7-120 bin]# ./startup.sh
Using CATALINA_BASE:   /opt/apache-tomcat-9.0.121
Using CATALINA_HOME:   /opt/apache-tomcat-9.0.121
Using CATALINA_TMPDIR: /opt/apache-tomcat-9.0.121/temp
Using JRE_HOME:        /usr
Using CLASSPATH:       /opt/apache-tomcat-9.0.121/bin/bootstrap.jar:/opt/apache-tomcat-9.0.121/bin/tomcat-juli.jar
Tomcat started.
[root@ip-172-31-7-120 bin]#
```

- Opened browser
- Accessed:
- `http://<tomcat-public-ip>:8080`
- Apache Tomcat welcome page displayed
- Confirms Tomcat is running successfully



- Searched for context.xml files:
- `find /opt/apache-tomcat-9.0.113-name context.xml`
- Identified multiple context.xml locations
- Required for manager access configuration

```
[root@ip-172-31-7-120 bin]# find / -name context.xml
/opt/apache-tomcat-9.0.121/conf/context.xml
/opt/apache-tomcat-9.0.121/webapps/docs/META-INF/context.xml
/opt/apache-tomcat-9.0.121/webapps/examples/META-INF/context.xml
/opt/apache-tomcat-9.0.121/webapps/host-manager/META-INF/context.xml
/opt/apache-tomcat-9.0.121/webapps/manager/META-INF/context.xml
[root@ip-172-31-7-120 bin]# vi /opt/apache-tomcat-9.0.121/webapps/docs/META-INF/context.xml
```

- Edited context.xml
- Commented or modified **RemoteAddrValve**:
- `<!-- <Valve className="org.apache.catalina.valves.RemoteAddrValve"`
- `allow="127\.\0\.\0\.\1|::1|.*" /> -->`
- Allows Jenkins server to deploy remotely
- Saved configuration

```
aws
[Alt+S]
IAM VPC EC2 S3 Route 53 Certificate Manager
United States (N. Virginia) Srihari

?xml version="1.0" encoding="UTF-8"?>
<!--
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contributor license agreements. See the NOTICE file distributed with
this work for additional information regarding copyright ownership.
The ASF licenses this file to You under the Apache License, Version 2.0
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distributed under the License is distributed on an "AS IS" BASIS,
WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
See the License for the specific language governing permissions and
limitations under the License.
-->
<Context antiResourceLocking="false" >
    <!-- <Valve className="org.apache.catalina.valves.RemoteCIDRValve"
        allow="127.0.0.0/8,::1/128" /> -->
</Context>

[root@ip-172-31-7-120 bin]# find / -name context.xml
/opt/apache-tomcat-9.0.121/conf/context.xml
/opt/apache-tomcat-9.0.121/webapps/docs/META-INF/context.xml
/opt/apache-tomcat-9.0.121/webapps/examples/META-INF/context.xml
/opt/apache-tomcat-9.0.121/webapps/host-manager/META-INF/context.xml
/opt/apache-tomcat-9.0.121/webapps/manager/META-INF/context.xml
[root@ip-172-31-7-120 bin]# vi /opt/apache-tomcat-9.0.121/webapps/docs/META-INF/context.xml
[root@ip-172-31-7-120 bin]# vi /opt/apache-tomcat-9.0.121/webapps/examples/META-INF/context.xml

<?xml version="1.0" encoding="UTF-8"?>
<!--
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contributor license agreements. See the NOTICE file distributed with
this work for additional information regarding copyright ownership.
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the License. You may obtain a copy of the License at

    http://www.apache.org/licenses/LICENSE-2.0

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distributed under the License is distributed on an "AS IS" BASIS,
WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
See the License for the specific language governing permissions and
limitations under the License.
-->
<Context>
    <CookieProcessor className="org.apache.tomcat.util.http.Rfc6265CookieProcessor"
        sameSiteCookies="strict" />
    <!-- <Valve className="org.apache.catalina.valves.RemoteCIDRValve"
        allow="127.0.0.0/8,::1/128" /> -->
</Context>

[root@ip-172-31-7-120 bin]# find / -name context.xml
/opt/apache-tomcat-9.0.121/conf/context.xml
/opt/apache-tomcat-9.0.121/webapps/docs/META-INF/context.xml
/opt/apache-tomcat-9.0.121/webapps/examples/META-INF/context.xml
/opt/apache-tomcat-9.0.121/webapps/host-manager/META-INF/context.xml
/opt/apache-tomcat-9.0.121/webapps/manager/META-INF/context.xml
[root@ip-172-31-7-120 bin]# vi /opt/apache-tomcat-9.0.121/webapps/docs/META-INF/context.xml
[root@ip-172-31-7-120 bin]# vi /opt/apache-tomcat-9.0.121/webapps/examples/META-INF/context.xml
[root@ip-172-31-7-120 bin]# vi /opt/apache-tomcat-9.0.121/webapps/host-manager/META-INF/context.xml

<?xml version="1.0" encoding="UTF-8"?>
<!--
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contributor license agreements. See the NOTICE file distributed with
this work for additional information regarding copyright ownership.
The ASF licenses this file to You under the Apache License, Version 2.0
(the "License"); you may not use this file except in compliance with
the License. You may obtain a copy of the License at

    http://www.apache.org/licenses/LICENSE-2.0

Unless required by applicable law or agreed to in writing, software
distributed under the License is distributed on an "AS IS" BASIS,
WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
See the License for the specific language governing permissions and
limitations under the License.
-->
<Context antiResourceLocking="false" privileged="true" >
    <CookieProcessor className="org.apache.tomcat.util.http.Rfc6265CookieProcessor"
        sameSiteCookies="strict" />
    <!-- <Valve className="org.apache.catalina.valves.RemoteCIDRValve"
        allow="127.0.0.0/8,::1/128" /> -->
    <Manager sessionAttributeValueClassNameFilter="java\\.lang\\. (?:Boolean|Integer|Long|Number|String) |org\\.apache\\.catalina\\.filters\\.CsrfPreventionFilter"
        ?(java\\.util\\. (?Linked)?HashMap)" />
</Context>

-- INSERT --

22,37

[root@ip-172-31-7-120 bin]# find / -name context.xml
/opt/apache-tomcat-9.0.121/conf/context.xml
/opt/apache-tomcat-9.0.121/webapps/docs/META-INF/context.xml
/opt/apache-tomcat-9.0.121/webapps/examples/META-INF/context.xml
/opt/apache-tomcat-9.0.121/webapps/host-manager/META-INF/context.xml
/opt/apache-tomcat-9.0.121/webapps/manager/META-INF/context.xml
[root@ip-172-31-7-120 bin]# vi /opt/apache-tomcat-9.0.121/webapps/docs/META-INF/context.xml
[root@ip-172-31-7-120 bin]# vi /opt/apache-tomcat-9.0.121/webapps/examples/META-INF/context.xml
[root@ip-172-31-7-120 bin]# vi /opt/apache-tomcat-9.0.121/webapps/host-manager/META-INF/context.xml
[root@ip-172-31-7-120 bin]# vi /opt/apache-tomcat-9.0.121/webapps/manager/META-INF/context.xml
```

```
<?xml version="1.0" encoding="UTF-8"?>
<!--
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contributor license agreements. See the NOTICE file distributed with
this work for additional information regarding copyright ownership.
The ASF licenses this file to You under the Apache License, Version 2.0
(the "License"); you may not use this file except in compliance with
the License. You may obtain a copy of the License at

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distributed under the License is distributed on an "AS IS" BASIS,
WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
See the License for the specific language governing permissions and
limitations under the License.
-->
<Context antiResourceLocking="false" privileged="true" >
    <CookieProcessor className="org.apache.tomcat.util.http.Rfc6265CookieProcessor"
        sameSiteCookies="strict" />
    <!-- <Valve className="org.apache.catalina.valves.RemoteCIDRValve"
        allow="127.0.0.0/8,::1/128" /> -->
    <Manager sessionAttributeValueTypeClassNameFilter="java.lang. (?Boolean|Integer|Long|Number|String) |org.apache.catalina.filters.CsrfPreventionFilter|$LruCache(?:
s1)?|java.util. (?|Linked)?HashMap"/>
</Context>
```

Opened:

- vi tomcat-users.xml
- Prepared to configure users and roles for manager access.
- Added roles:
- <role rolename="manager-gui"/>
- <role rolename="manager-script"/>
- <role rolename="manager-status"/>
- Added user for Jenkins:
- <user username="deployer" password="deployer"
- roles="manager-gui,manager-script,manager-status"/>
- Saved the file.

```
[root@ip-172-31-7-120 ~]# cd /opt/apache-tomcat-9.0.121/conf/
[root@ip-172-31-7-120 conf]# ls
catalina catalina.properties jaspic-providers.xml logging.properties tomcat-users.xml web.xml
catalina.policy context.xml jaspic-providers.xsd server.xml tomcat-users.xsd
[root@ip-172-31-7-120 conf]# vi tomcat-users.xml
[root@ip-172-31-7-120 conf]#
```

application, do not forget to remove the <!-- ... --> that surrounds them. You will also need to set the passwords to something appropriate.

```
-->
<!--
<user username="admin" password="<must-be-changed>" roles="manager-gui"/>
<user username="robot" password="<must-be-changed>" roles="manager-script"/>
-->
<!--
The sample user and role entries below are intended for use with the
examples web application. They are wrapped in a comment and thus are ignored
when reading this file. If you wish to configure these users for use with the
examples web application, do not forget to remove the <!-- ... --> that surrounds
them. You will also need to set the passwords to something appropriate.
-->
<!--
<role rolename="tomcat"/>
<role rolename="role1"/>
<user username="tomcat" password="<must-be-changed>" roles="tomcat"/>
<user username="both" password="<must-be-changed>" roles="tomcat,role1"/>
<user username="role1" password="<must-be-changed>" roles="role1"/>
-->
<role rolename="manager-gui"/>
<role rolename="manager-script"/>
<role rolename="manager-jmx"/>
<role rolename="manager-status"/>
<user username="admin" password="admin" roles="manager-gui, manager-script, manager-jmx, manager-status"/>
<user username="deployer" password="deployer" roles="manager-script"/>
<user username="tomcat" password="s3cret" roles="manager-gui"/>
</tomcat-users>
-- INSERT --
```

62, 65-72

i-Oa8056acfe3f4019f (NEXUS-SERVER)

PublicIPs: 100.24.106.233 PrivateIPs: 172.31.7.120

Restarted Tomcat to apply changes:

- cd /opt/apache-tomcat-9.0.113/bin
- ./shutdown.sh
- ./startup.sh
- Tomcat started successfully with updated access rules

```
[root@ip-172-31-7-120 conf]# cd ..
[root@ip-172-31-7-120 apache-tomcat-9.0.121]# cd bin/
[root@ip-172-31-7-120 bin]# ls
bootstrap.jar      catalina.sh        commons-daemon-native.tar.gz  configtest.sh  digest.sh          setclasspath.bat  shutdown.sh  tomcat-juli.jar  tool-wrapper.sh
catalina-tasks.xml  ciphers.bat        commons-daemon.jar           daemon.sh       makebase.bat      setclasspath.sh  startup.bat  tomcat-native.tar.gz  version.bat
catalina.bat        configtest.bat     digest.bat                   makebase.sh    shutdown.bat      startup.sh        tool-wrapper.bat  version.sh
[root@ip-172-31-7-120 bin]# ./shutdown.sh
Using CATALINA_BASE:   /opt/apache-tomcat-9.0.121
Using CATALINA_HOME:   /opt/apache-tomcat-9.0.121
Using CATALINA_TMPDIR: /opt/apache-tomcat-9.0.121/temp
Using JRE_HOME:        /usr
Using CLASSPATH:       /opt/apache-tomcat-9.0.121/bin/bootstrap.jar:/opt/apache-tomcat-9.0.121/bin/tomcat-juli.jar
NOTE: Picked up JDK_JAVA_OPTIONS:  --add-opens=java.base/java.lang=ALL-UNNAMED --add-opens=java.base/java.lang.invoke=ALL-UNNAMED --add-opens=java.base/java.lang.reflect=ALL-UNNAMED --add-opens=java.base/java.io=ALL-UNNAMED --add-opens=java.base/java.util=ALL-UNNAMED --add-opens=java.base/java.util.concurrent=ALL-UNNAMED --add-opens=java.rmi/sun.rmi.transport=ALL-UNNAMED
[root@ip-172-31-7-120 bin]# ./startup.sh
Using CATALINA_BASE:   /opt/apache-tomcat-9.0.121
Using CATALINA_HOME:   /opt/apache-tomcat-9.0.121
Using CATALINA_TMPDIR: /opt/apache-tomcat-9.0.121/temp
Using JRE_HOME:        /usr
Using CLASSPATH:       /opt/apache-tomcat-9.0.121/bin/bootstrap.jar:/opt/apache-tomcat-9.0.121/bin/tomcat-juli.jar
Using CATALINA_OPTS:
Tomcat started.
[root@ip-172-31-7-120 bin]#
```

i-Oa8056acfe3f4019f (NEXUS-SERVER)

PublicIP: 100.24.106.233 PrivateIP: 172.31.7.120

100.24.106.233:8080/manager/status

Sign in to access this site

Authorization required by http://100.24.106.233:8080
Your connection to this site is not secure

Username

Password



Server Status

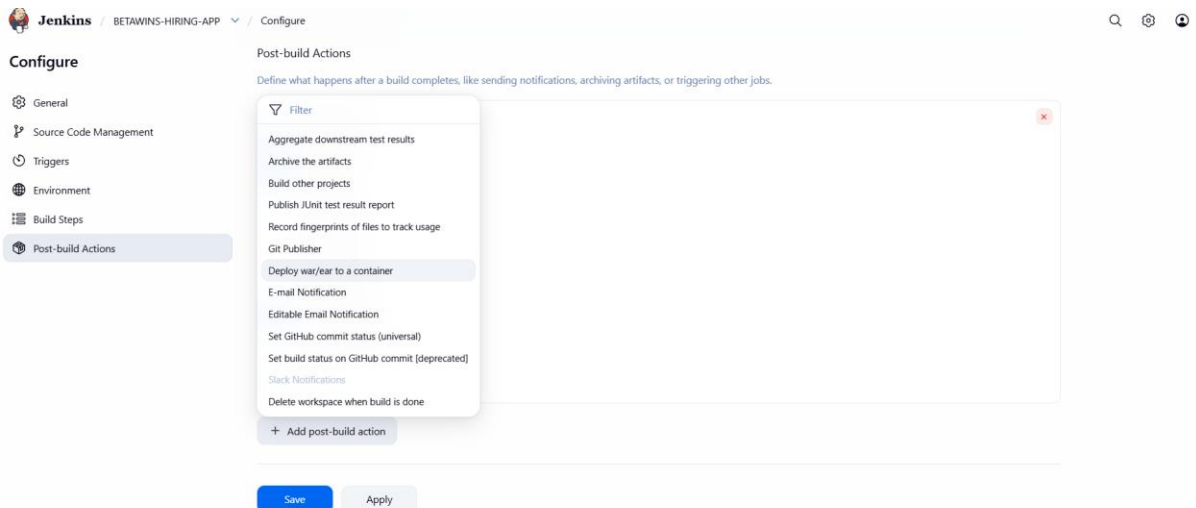
Manager			
List Applications	HTML Manager Help	Manager Help	Complete Server Status

Server Information							
Tomcat Version	JVM Version	JVM Vendor	OS Name	OS Version	OS Architecture	Hostname	IP Address
Apache Tomcat/9.0.121	21.0.12+8-LTS	Amazon.com Inc.	Linux	6.18.41-94.142.amzn2023.x86_64	amd64	ip-172-31-7-120.ec2.internal	172.31.7.120

JVM						
Free Memory: 39.58 MiB Total Memory: 60.00 MiB Max Memory: 954.00 MiB						
Memory Pool	Type	Initial	Total	Maximum	Used	
G1 Eden Space	Heap memory	6.00 MiB	36.00 MiB	-0.00 MiB	6.00 MiB	
G1 Old Gen	Heap memory	54.00 MiB	22.00 MiB	954.00 MiB	12.30 MiB (1%)	
G1 Survivor Space	Heap memory	0.00 MiB	2.00 MiB	-0.00 MiB	1.16 MiB	
CodeHeap 'non-nmethods'	Non-heap memory	2.43 MiB	2.43 MiB	5.55 MiB	1.58 MiB (28%)	
CodeHeap 'non-profiled nmethods'	Non-heap memory	2.43 MiB	2.43 MiB	117.22 MiB	1.49 MiB (1%)	
CodeHeap 'profiled nmethods'	Non-heap memory	2.43 MiB	7.25 MiB	117.21 MiB	7.22 MiB (6%)	
Compressed Class Space	Non-heap memory	0.00 MiB	2.56 MiB	1024.00 MiB	2.39 MiB (0%)	
Metaspace	Non-heap memory	0.00 MiB	25.43 MiB	-0.00 MiB	25.10 MiB	

"http-nio-8080"	
Max threads: 200 Current thread count: 10 Current threads busy: 1 Keep alive sockets count: 1	
Max processing time: 712 ms Processing time: 0.874 s Request count: 4 Error count: 1 Bytes received: 0.00 MiB Bytes sent: 0.03 MiB	

- Open Jenkins dashboard
- Select job **BETAWINS-HIRING-APP**
- Click **Configure**
- Scroll to **Post-build Actions**
- Click **Add post-build action**
- Select **Deploy war/ear to a container**
- In **WAR/EAR files** field, enter: ****/*.war**
- Under **Containers**
- Click **Add Container**
- Select **Tomcat 9.x Remote**



Jenkins / BETAWINS-HIRING-APP / Configure

Configure

- General
- Source Code Management
- Triggers
- Environment
- Build Steps
- Post-build Actions**

Post-build Actions

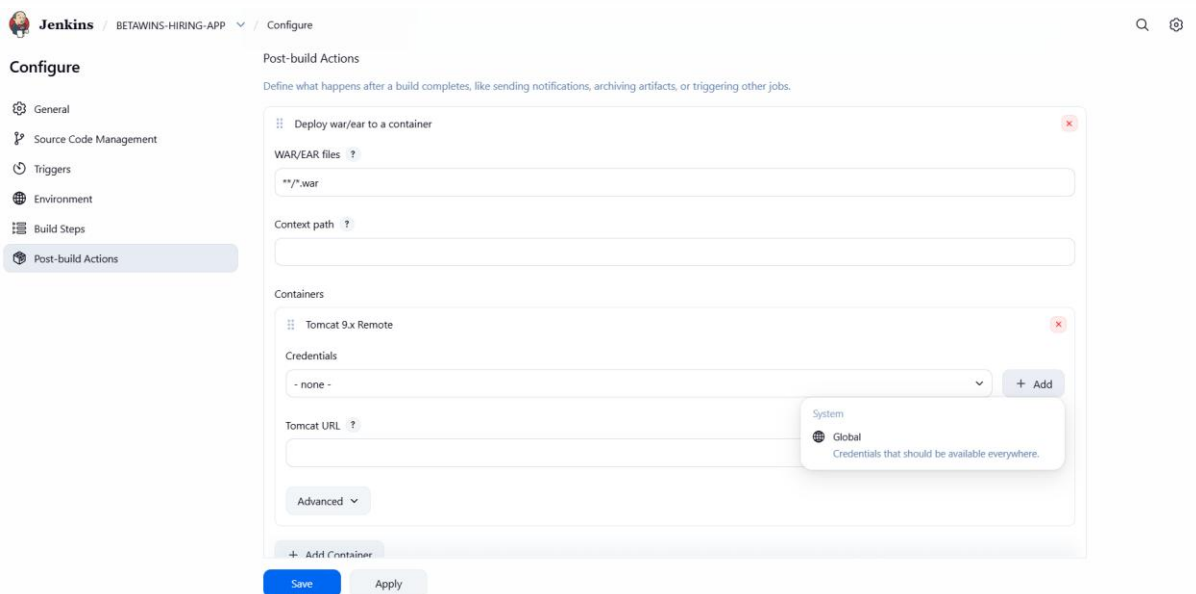
Define what happens after a build completes, like sending notifications, archiving artifacts, or triggering other jobs.

Filter

- Aggregate downstream test results
- Archive the artifacts
- Build other projects
- Publish JUnit test result report
- Record fingerprints of files to track usage
- Git Publisher
- Deploy war/ear to a container**
- E-mail Notification
- Editable Email Notification
- Set GitHub commit status (universal)
- Set build status on GitHub commit (deprecated)
- Slack Notifications
- Delete workspace when build is done

+ Add post-build action

Save Apply



Jenkins / BETAWINS-HIRING-APP / Configure

Configure

- General
- Source Code Management
- Triggers
- Environment
- Build Steps
- Post-build Actions**

Post-build Actions

Define what happens after a build completes, like sending notifications, archiving artifacts, or triggering other jobs.

Deploy war/ear to a container

WAR/EAR files ?

**/*.war

Context path ?

Containers

Tomcat 9.x Remote

Credentials

- none - + Add

Tomcat URL ?

Advanced

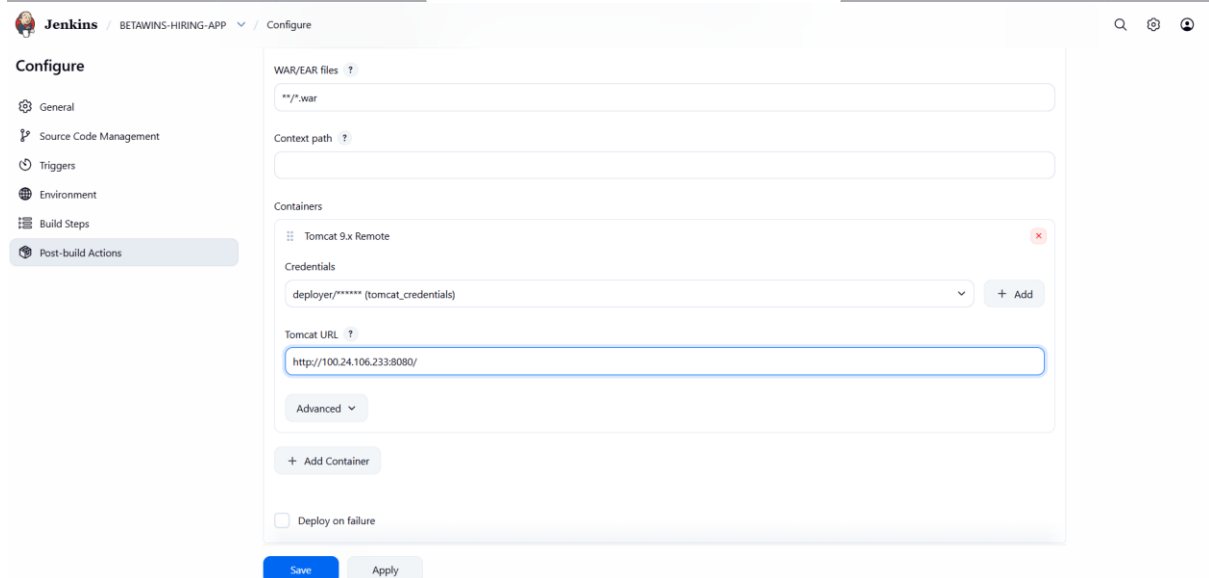
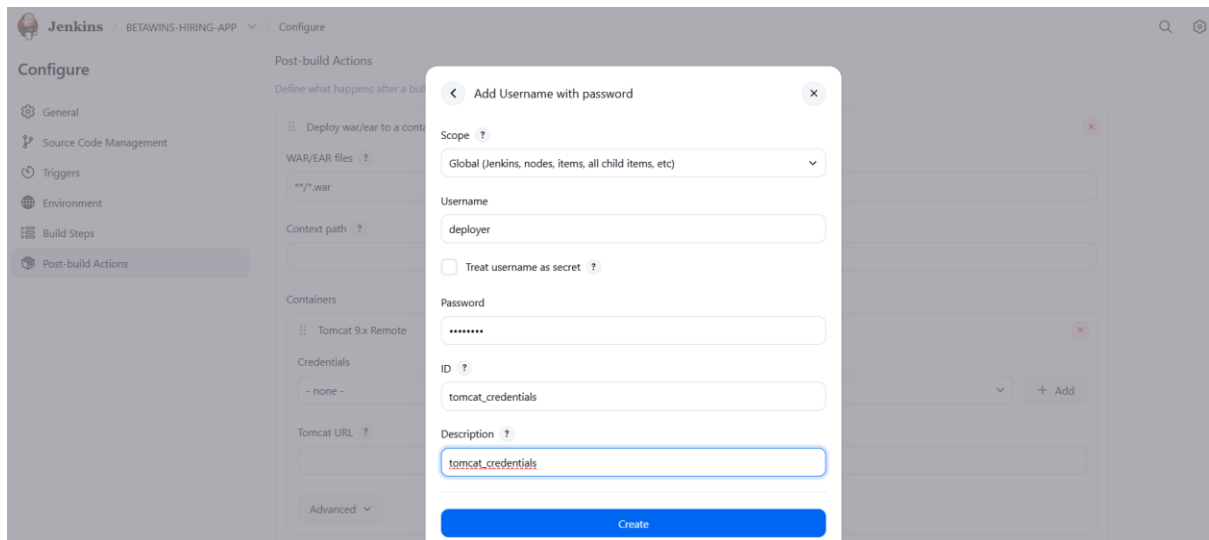
+ Add Container

Save Apply

System

Global

Credentials that should be available everywhere.



Jenkins / BETAWINS-HIRING-APP / #11

Status

Changes

Console Output

Delete build '#11'

Git Build Data

Edit build information

Previous Build

#11 (Aug 18, 2026, 2:53:18 PM)

Started by user BOORA SRIHARI

Revision: 9405b107dfdf68e69fe953ea1f9669784f9d6a
Repository: <https://github.com/betawins/hiring-app.git>

refs/remotes/origin/main

No changes.

Add description

Keep this build forever

Started 28 sec ago
Took 15 sec on built-in

```
[INFO] Installing /var/lib/jenkins/workspace/BETAWINS-HIRING-APP/pom.xml to /var/lib/jenkins/.m2/repository/in/javahome/hiring/v.1/hiring-v.1.pom
[INFO] Installing /var/lib/jenkins/workspace/BETAWINS-HIRING-APP/target/hiring.war to /var/lib/jenkins/.m2/repository/in/javahome/hiring/0.1/hiring-0.1.war
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 1.437 s
[INFO] Finished at: 2026-08-18T14:53:34Z
[INFO] -----
Uploading artifact hiring.war started....
GroupId: in.javahome
ArtifactId: hiring
Classifier:
Type: war
Version: 0.1
File: hiring.war
Repository:maven-releases-01
Uploading: http://100.24.106.233:8081/repository/maven-releases-01/content/repositories/maven-releases-01/in/javahome/hiring/0.1/hiring-0.1.war
100 % completed (1.7 kB / 1.7 kB).
Uploaded: http://100.24.106.233:8081/repository/maven-releases-01/content/repositories/maven-releases-01/in/javahome/hiring/0.1/hiring-0.1.war (1.7 kB at 29 kB/s)
Uploading artifact hiring.war completed.
[DeployPublisher][INFO] Attempting to deploy 1 war file(s)
[DeployPublisher][INFO] Deploying /var/lib/jenkins/workspace/BETAWINS-HIRING-APP/target/hiring.war to container Tomcat 9.x Remote with context null
[/var/lib/jenkins/workspace/BETAWINS-HIRING-APP/target/hiring.war] is not deployed. Doing a fresh deployment.
Deploying [/var/lib/jenkins/workspace/BETAWINS-HIRING-APP/target/hiring.war]
[Slack Notifications] found #10 as previous completed, non-aborted build
[Slack Notifications] will send OnSuccessNotification because build matches and user preferences allow it
Finished: SUCCESS
```

Verify WAR Deployment on Server

- SSH into Tomcat server

```
cd /opt/apache-tomcat-9.0.113/webapps
```

```
ls
```

- hiring folder and hiring.war are present

```
[ec2-user@ip-172-31-7-120 ~]$ sudo su
[root@ip-172-31-7-120 ec2-user]# cd /opt/apache-tomcat-9.0.121/
[root@ip-172-31-7-120 apache-tomcat-9.0.121]# cd webapps/
[root@ip-172-31-7-120 webapps]# ls
ROOT  docs  examples  hiring  hiring.war  host-manager  manager
[root@ip-172-31-7-120 webapps]#
```

EC2 Instance Co... x EC2 Instance Co... x BETAWINS-HIR... x Credentials - Mu... x Browse - Sonat... x build_notificati... x /manager x 100.24.106.233:8080 x

Not secure 100.24.106.233:8080/hiring/

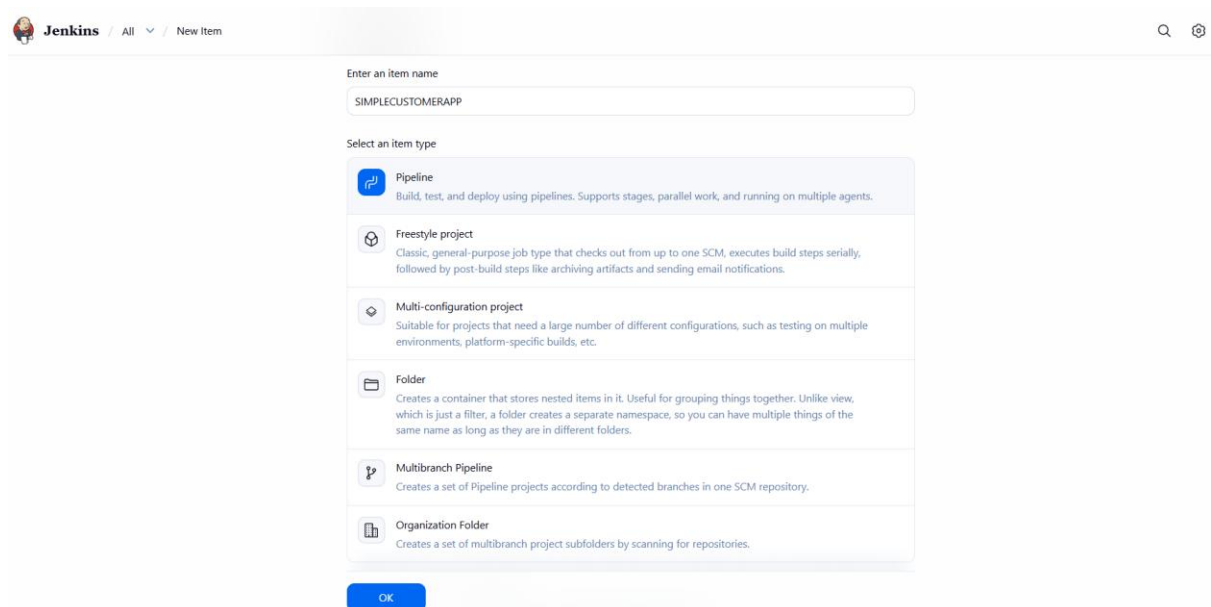
Welcome to Techie Horizon

2) Setup a Jenkins CICD pipeline using Declarative pipeline using feature-1.1 branch.

Source Code: https://github.com/betawins/sabear_simplecutomerapp/tree/feature-1.1
stages:

- Git Clone
- SonarQube Integration
- Maven Compilation
- Nexus Artifactory
- Slack Notification
- Deploy On tomcat

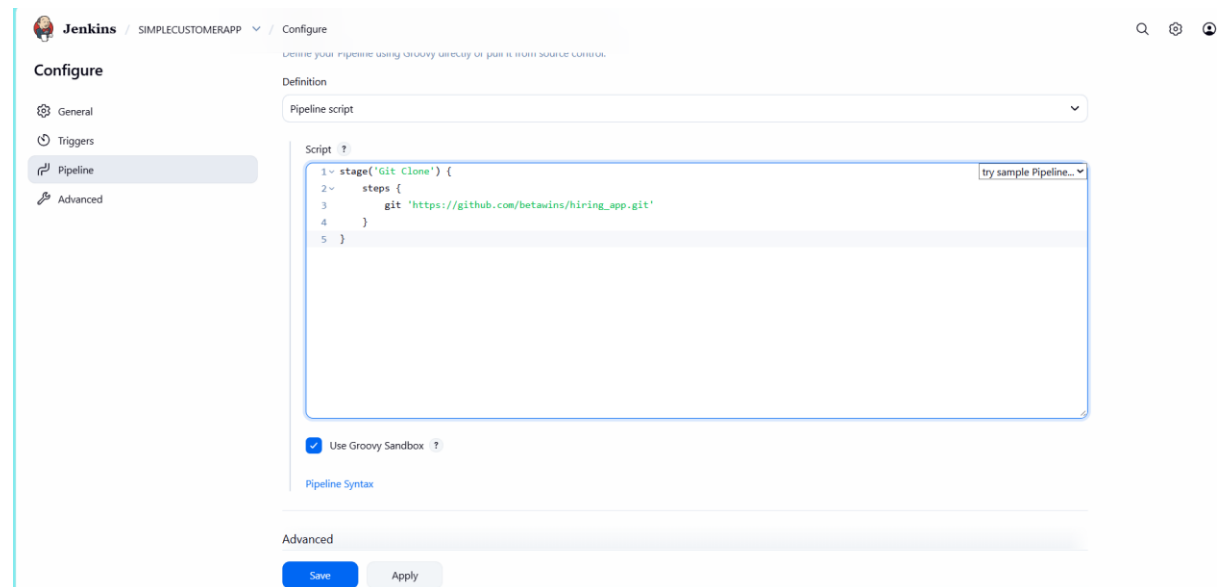
- Opened **Jenkins** dashboard
- Clicked **New Item**
- Entered job name: **SIMPLECUSTOMERAPP**
- Selected **Pipeline** as the job type
- Clicked **OK**



Opened job Configure

- Navigated to Pipeline section
- Selected Pipeline script
- Added basic pipeline code with:

- agent any
- Single stage Git Clone
- GitHub repository URL
- Clicked Save



- Clicked **Build Now**
- Pipeline executed **Git Clone** stage
- Build **failed**

Jenkins / SIMPLECUSTOMERAPP / #1

Status ✖ #1 (Aug 18, 2026, 5:27:03 PM) 🔍 ⚙️ 👤

Changes 🕒 Started by user BOORA SRIHARI Started 43 sec ago

Console Output 🔄 No changes. Took 0.63 sec

Delete build #1

Pipeline Overview

Edit build information

Replay

Pipeline Steps

Workspaces

Add description Keep this build forever

Jenkins / SIMPLECUSTOMERAPP / Configure 🔍 ⚙️

Configure

General

Triggers

Pipeline

Advanced

Definition

Pipeline script

```

1 pipeline {
2   agent {
3     label 'master'
4   }
5
6   stages {
7     stage('Git Clone') {
8       steps {
9         git 'https://github.com/betawins/sabear_simplecoterapp.git'
10      }
11    }
12  }
13 }

```

☒ Use Groovy Sandbox ?

[Pipeline Syntax](#)

Advanced

Jenkins / SIMPLECUSTOMERAPP / #6 🔍

Running on Jenkins in /var/lib/jenkins/workspace/SIMPLECUSTOMERAPP

Git Build Data

Pipeline Overview

Edit build information

Restart from Stage

Replay

Pipeline Steps

Workspaces

Previous Build

```

[Pipeline] {
[Pipeline] stage
[Pipeline] { (Git Clone)
[Pipeline] git
The recommended git tool is: NONE
No credentials specified
Cloning the remote Git repository
Cloning repository https://github.com/betawins/sabear_simplecoterapp.git
> git init /var/lib/jenkins/workspace/SIMPLECUSTOMERAPP # timeout=10
Fetching upstream changes from https://github.com/betawins/sabear_simplecoterapp.git
> git --version # timeout=10
> git --version # 'git version 2.50.1'
> git fetch --tags --force --progress -- https://github.com/betawins/sabear_simplecoterapp.git +refs/heads/*:refs/remotes/origin/* # timeout=10
> git config remote.origin.url https://github.com/betawins/sabear_simplecoterapp.git # timeout=10
> git config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeout=10
Avoid second fetch
> git rev-parse refs/remotes/origin/master^{commit} # timeout=10
Checking out Revision 9be96d5661cf61f49c0b8d076dc20d2b09624f4e (refs/remotes/origin/master)
> git config core.sparsecheckout # timeout=10
> git checkout -f 9be96d5661cf61f49c0b8d076dc20d2b09624f4e # timeout=10
> git branch -a -v --no-abbrev # timeout=10
> git checkout -b master 9be96d5661cf61f49c0b8d076dc20d2b09624f4e # timeout=10
Commit message: "Update README.md"
First time build. Skipping changelog.
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS

```

- Updated pipeline to include **SonarQube Analysis** stage
- Used:
 - withSonarQubeEnv('sonarqube-server')
 - Maven command with Sonar parameters
- Saved pipeline configuration

Configure

- General
- Triggers
- Pipeline
- Advanced

Pipeline

Define your Pipeline using Groovy directly or pull it from source control.

Definition

Pipeline script

Script ?

```
12
13 stage('SonarQube Analysis') {
14     steps {
15         withSonarQubeEnv('SonarQube') {
16             sh '''
17                 mvn clean verify \
18                     org.sonarsource.scanner.maven:sonar-maven-plugin:sonar \
19                     -Dsonar.projectKey=SimpleCustomerApp \
20                     -Dsonar.projectName=SimpleCustomerApp
21             '''
22         }
23     }
24 }
25
26 }
```

☒ Use Groovy Sandbox ?

- Pipeline started successfully
- Git Clone** stage passed
- SonarQube Analysis** stage failed
- Error highlighted in red in stage view



Jenkins / SIMPLECUSTOMERAPP

Status

</> Changes

► Build Now

🗑 Delete Pipeline

⚙ Configure

✎ Rename

🔊 SonarQube

📁 Stages

❓ Pipeline Syntax

✓ SIMPLECUSTOMERAPP

SonarQube Quality Gate

SimpleCustomerApp **Passed**

server-side processing: **Success**

Permalinks

- Last build (#11), 4 min 26 sec ago
- Last stable build (#11), 4 min 26 sec ago
- Last successful build (#11), 4 min 26 sec ago
- Last failed build (#10), 8 min 48 sec ago
- Last unsuccessful build (#10), 8 min 48 sec ago
- Last completed build (#11), 4 min 26 sec ago

Builds >

Filter

/

Today

✓

#11

7:09 PM

✗

#10

7:04 PM

✗

#9

6:44 PM

Jenkins / SIMPLECUSTOMERAPP #11

```
[INFO] SCM Publisher 1 source file to be analyzed
[INFO] SCM Publisher 1/1 source file have been analyzed (done) | time=240ms
[INFO] CPD Executor Calculating CPD for 0 files
[INFO] CPD Executor CPD calculation finished (done) | time=0ms
[INFO] SCM revision ID '9be96d5661cf61f49c0b8d076dc20d2b09624f4e'
[INFO] Analysis report generated in 192ms, dir size=288.6 kB
[INFO] Analysis report compressed in 29ms, zip size=31.2 kB
[INFO] Analysis report uploaded in 309ms
[INFO] ANALYSIS SUCCESSFUL, you can find the results at: http://32.197.36.115:9000/dashboard?id=SimpleCustomerApp
[INFO] Note that you will be able to access the updated dashboard once the server has processed the submitted analysis report
[INFO] More about the report processing at http://32.197.36.115:9000/api/ce/task?id=574f785d-c82b-4eb7-a149-ef3384c314f8
[INFO] Analysis total time: 3.111 s
[INFO] SonarScanner Engine completed successfully
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 11.139 s
[INFO] Finished at: 2026-08-18T19:09:28Z
[INFO] -----
[Pipeline] }
[Pipeline] // withSonarQubeEnv
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

The way in which security, reliability, and maintainability counts and ratings are calculated has changed. [Learn more in SonarQube documentation](#)

SonarQube community

Projects Issues Rules Quality profiles Quality gates Administration More

My Favorites All

Filters

Quality gate

Passed 2

Failed 0

Security

Create Project

Search projects Perspective Overall Status Sort by Name 2 project(s)

SimpleCustomerApp Public Passed

Last analysis: 6 minutes ago - 28 Lines of Code - XML

A 0 A 0 A 0 A — — 0.0%

Security Reliability Maintainability Hotspots Reviewed Coverage Duplications

Opened **Jenkins** → **Job** → **Configure**

Defined a **Declarative Pipeline**

Added agent any

Configured **environment variables**:

- Maven home
- PATH
- Sonar project key and name

Defined stages:

- Git Clone
- SonarQube Analysis
- Maven Build

Saved the pipeline configuration

Configure

- General
- Triggers
- Pipeline
- Advanced

Script ?

```
20         -Dsonar.projectKey=SimpleCustomerApp \
21         -Dsonar.projectName=SimpleCustomerApp
22         ...
23     }
24 }
25 }
26
27 stage('Maven Compilation') {
28     steps {
29         sh 'mvn clean package -DskipTests'
30     }
31 }
32
33 }
34 }
```

☒ Use Groovy Sandbox ?

[Pipeline Syntax](#)



Jenkins / SIMPLECUSTOMERAPP #12



```
[INFO] --- resources:3.4.0:testResources (default-testResources) @ SimpleCustomerApp ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /var/lib/jenkins/workspace/SIMPLECUSTOMERAPP/src/test/resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ SimpleCustomerApp ---
[INFO] No sources to compile
[INFO]
[INFO] --- surefire:3.5.4:test (default-test) @ SimpleCustomerApp ---
[INFO] Tests are skipped.
[INFO]
[INFO] --- war:3.3.2:war (default-war) @ SimpleCustomerApp ---
[INFO] Packaging webapp
[INFO] Assembling webapp [SimpleCustomerApp] in [/var/lib/jenkins/workspace/SIMPLECUSTOMERAPP/target/SimpleCustomerApp-12-SNAPSHOT]
[INFO] Processing war project
[INFO] Building war: /var/lib/jenkins/workspace/SIMPLECUSTOMERAPP/target/SimpleCustomerApp-12-SNAPSHOT.war
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 1.405 s
[INFO] Finished at: 2026-08-18T19:19:11Z
[INFO] -----
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

```
[root@ip-172-31-4-173 ec2-user]# cd /var/lib/jenkins/workspace/
[root@ip-172-31-4-173 workspace]# ls
BETAWINS-HIRING-APP  SIMPLECUSTOMERAPP@tmp  jenkins  sample-1  sample-3  stage_2  webhook@tmp
SIMPLECUSTOMERAPP  first-job-01          jenkins@tmp  sample-2  stage_1  webhook
[root@ip-172-31-4-173 workspace]# cd SIMPLECUSTOMERAPP/
[root@ip-172-31-4-173 SIMPLECUSTOMERAPP]# ls -lh target/
total 4.0K
-rwxr-xr-x. 4 jenkins jenkins 37 Aug 18 19:19 SimpleCustomerApp-12-SNAPSHOT
-rw-r--r--. 1 jenkins jenkins 1.4K Aug 18 19:19 SimpleCustomerApp-12-SNAPSHOT.war
-rwxr-xr-x. 2 jenkins jenkins 28 Aug 18 19:19 maven-archiver
[root@ip-172-31-4-173 SIMPLECUSTOMERAPP]#
```

sonatype nexus repository

Community Edition

Q Search components or CVEs...

🔔

🔗

⚙️

👤

Dashboard

Search

Browse

Upload

Malware Risk

Settings

Settings

Repository

Repositories

Blob Stores

Data Store

Proprietary Repositories

Content Selectors

Cleanup Policies

Routing Rules

Security

Support

System

IQ Server

Repositories / Select Recipe / Create Repository: maven2 (hosted)

Name: A unique identifier for this repository
simpleCustomerApp

Online: ☒ If checked, the repository accepts incoming requests

Maven 2

Version policy: What type of artifacts does this repository store?
Snapshot

Layout policy: Validate that all paths are maven artifact or metadata paths
Permissive

Content Disposition: Add Content-Disposition header as Attachment to disable some content from being inline in a browser.
Inline

⚠️ Serving content inline allows uploaded HTML to render on a trusted Nexus URL, which can be exploited for phishing attacks against other users.

Storage

Blob store: Blob store used to store repository contents
default

Strict Content Type Validation: ☒ Validate that all content uploaded to this repository is of a MIME type appropriate for the repository format

Hosted

Deployment policy: Controls if deployments of end updates to artifacts are allowed
Allow redeploy

Jenkins / SIMPLECUSTOMERAPP / Configure

🔍

⚙️

👤

Configure

General

Triggers

Pipeline

Advanced

41
42
43
44
45
46
47
48
49
50
51
52
53
54

```
stage('Nexus Upload') {  
  steps {  
    withCredentials([  
      usernamePassword(  
        credentialsId: 'nexus-credentials',  
        usernameVariable: 'NEXUS_USER',  
        passwordVariable: 'NEXUS_PASSWORD'  
      )  
    ]) {  
      sh '''  
        WAR_FILE=$(ls target/*.war)  
        echo "WAR file: $WAR_FILE"  
      '''  
    }  
  }  
}
```

☒ Use Groovy Sandbox

[Pipeline Syntax](#)

Jenkins
Manage Jenkins
Credentials
System
Global

Global

Credentials that should be available everywhere.

private_repo	arihan03-sh*****
slave-01	ec2-user
sonarqube-docker	sonarqube-docker
slack_notification_token	slack_notification_token
NEXUS_SERVER	admin*****
tomcat_credentials	deployer*****

Add Username with password

Scope ?

Global (Jenkins, nodes, items, all child items, etc)

Username

admin

☐ Treat username as secret ?

Password

ID ?

nexus-credentials

Description ?

nexus-credentials

Create

+ Add Credentials

Jenkins
SIMPLECUSTOMERAPP
#18

```

} [1422 bytes data]
* upload completely sent off: 1422 bytes
< HTTP/1.1 201 Created
< Server: Nexus/3.95.1-01 (COMMUNITY)
< X-Content-Type-Options: nosniff
< Content-Security-Policy: sandbox allow-forms allow-modals allow-popups allow-presentation allow-scripts allow-top-navigation
< X-XSS-Protection: 0
< Content-Length: 0
<

100 1422 0 0 100 1422 0 52177 --:--:-- --:--:-- --:--:-- 52666
* Connection #0 to host 100.24.106.233:8081 left intact
+ echo 'Nexus upload completed.'
Nexus upload completed.
[Pipeline] }
[Pipeline] // withCredentials
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] echo
CI Pipeline completed successfully.
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS

```

sonatype nexus repository
Community Edition

Search components or CVEs...

Browser / simpleCustomerApp

HTML View

SimpleCustomerApp-18-SNAPSHOT.war

Advanced search...

SimpleCustomerApp-18-SNAPSHOT.war

Delete asset

Content type	application/java-archive
File size	1.4 KB
Blob created	Wed Aug 19 2026 01:36:31 GMT+0530 (India Standard Time)
Blob updated	Wed Aug 19 2026 01:36:31 GMT+0530 (India Standard Time)
Last downloaded	has not been downloaded
Locally cached	true
Blob reference	default@6a2964c6-dccc-4b42-909e-14957a5293f@2026-08-18T20:06
Containing repo	simpleCustomerApp
Uploader	admin
Uploader's IP Address	32.195.69.15

Jenkins / SIMPLECUSTOMERAPP / Configure

Configure

General

Triggers

Pipeline

Advanced

```
68 stage('Slack Notification') {
69     steps {
70         slackSend(
71             channel: '#build_notification',
72             message: "SUCCESS: ${env.JOB_NAME} #${env.BUILD_NUMBER} - Build and Nexus upload completed successfully."
73         )
74     }
75 }
76 }
77
78 post {
79     failure {
80         slackSend(
```

☒ Use Groovy Sandbox ?

[Pipeline Syntax](#)

Jenkins / SIMPLECUSTOMERAPP

Status

Changes

Build Now

Delete Pipeline

Configure

Rename

SonarQube

Stages

Pipeline Syntax

✓ SIMPLECUSTOMERAPP

Add description

SonarQube Quality Gate

SimpleCustomerApp **Passed**

server-side processing: **Success**

Permalinks

- Last build (#20), 2 min 30 sec ago
- Last stable build (#20), 2 min 30 sec ago
- Last successful build (#20), 2 min 30 sec ago
- Last failed build (#19), 3 min 17 sec ago
- Last unsuccessful build (#19), 3 min 17 sec ago
- Last completed build (#20), 2 min 30 sec ago

Builds >

Filter

Today

#20 8:18 PM

#19 8:18 PM

Jenkins / SIMPLECUSTOMERAPP / #20

```
>
} [1422 bytes data]
* upload completely sent off: 1422 bytes
< HTTP/1.1 201 Created
< Server: Nexus/3.95.1-01 (COMMUNITY)
< X-Content-Type-Options: nosniff
< Content-Security-Policy: sandbox allow-forms allow-modals allow-popups allow-presentation allow-scripts allow-top-navigation
< X-XSS-Protection: 0
< Content-Length: 0
<

100 1422 0 0 100 1422 0 60668 --:--:-- --:--:-- --:--:-- 61826
* Connection #0 to host 100.24.106.233:8081 left intact
+ echo 'Nexus upload completed.'
Nexus upload completed.
[Pipeline] }
[Pipeline] // withCredentials
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Slack Notification)
[Pipeline] slackSend
Slack Send Pipeline step running, values are - baseUrl: <empty>, teamDomain: hari, channel: #build_notification, color: <empty>, botUser: true, tokenCredentialI
slack_notification_token, notifyCommiters: false, iconEmoji: <empty>, username: <empty>, timestamp: <empty>
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Hari

Home

DMs

Activity

Files

More

Admin

build_notification

An offer awaits

Find a conversation...

Huddles

Drafts & sent

Directories

Starred

Drag and drop important stuff here

Channels

all-hari

build_notification

new-channel

social

+ Add channels

build_notification

Messages

BETAWINS-HIRING-APP - #10 Started by user BOORA SRIHARI

BETAWINS-HIRING-APP - #10 Success after 14 sec

Jenkins

BETAWINS-HIRING-APP - #11 Started by user BOORA SRIHARI

BETAWINS-HIRING-APP - #11 Success after 15 sec

Jenkins

Slack/Jenkins plugin: you're all set on <http://44.199.226.166:8081/>

FAILED: SIMPLECUSTOMERAPP #19 - Pipeline failed. Check Jenkins Console Output.

SUCCESS: SIMPLECUSTOMERAPP #20 - Build and Nexus upload completed successfully.

Jenkins

SIMPLECUSTOMERAPP

#22

Status

Changes

Console Output

Delete build '#22'

Git Build Data

Pipeline Overview

Edit build information

Restart from Stage

Replay

Pipeline Steps

Workspaces

Previous Build

#22 (Aug 18, 2026, 8:37:40 PM)

Started by user BOORA SRIHARI

Revision: 9be96d5661cf61f49c0b8d076dc20d2b09624f4e

Repository: https://github.com/betawins/sabear_simplecustomerapp.git

refs/remotes/origin/master

No changes.

Add description

Keep this build forever

Started 1 min 54 sec ago

Took 14 sec

```
[Pipeline] {
[Pipeline] sh
++ ls target/SimpleCustomerApp-22-SNAPSHOT.war
+ WAR_FILE=target/SimpleCustomerApp-22-SNAPSHOT.war
+ echo 'Deploying target/SimpleCustomerApp-22-SNAPSHOT.war to Tomcat...'
Deploying target/SimpleCustomerApp-22-SNAPSHOT.war to Tomcat...
+ curl -u ***** --upload-file target/SimpleCustomerApp-22-SNAPSHOT.war 'http://100.24.106.233:8080/manager/text/deploy?path=/SimpleCustomerApp&update=true'
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed

  0   0   0    0   0   0  --:--:-- --:--:-- --:--:--    0
100 1485   0  63 100 1422 2874 64890 --:--:-- --:--:-- --:--:-- 70714
OK - Deployed application at context path [/SimpleCustomerApp]
+ echo 'Deployment completed.'
Deployment completed.
[Pipeline] }
[Pipeline] // withCredentials
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] slackSend

Slack Send Pipeline step running, values are - baseUrl: <empty>, teamDomain: hari, channel: #build_notification, color: <empty>, botUser: true, tokenCredentialId:
slack_notification_token, notifyCommiters: false, iconEmoji: <empty>, username: <empty>, timestamp: <empty>

[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Pipeline stages now include:

- Notify Started
- Git Clone
- SonarQube Analysis
- Maven Build
- Nexus Upload
- Declarative: Post Actions

All stages completed successfully (green)

Slack notification triggered at the end

Jenkins / SIMPLECUSTOMERAPP
🔍 ⚙️ 👤

Status

[🔄 Changes](#)
[🏗 Build Now](#)
[🗑 Delete Pipeline](#)
[⚙ Configure](#)
[✎ Rename](#)
[🔍 Full Stage View](#)
[📊 SonarQube](#)
[📅 Stages](#)
[📘 Pipeline Syntax](#)

[Add description](#)

SIMPLECUSTOMERAPP

Stage View

	Git Clone	SonarQube Analysis	Maven Compilation	Nexus Upload	Slack Notification	Deploy On Tomcat	Declarative: Post Actions
Average stage times: (full run time: ~14s)	451ms	9s	2s	358ms	130ms	374ms	162ms
#22 Aug 19 02:07 No Changes	503ms	9s	2s	359ms	154ms	360ms	132ms
#21 Aug 19 02:02 No Changes	220ms	9s	2s	358ms	102ms	388ms failed	193ms
#20 Aug 19 01:48 No Changes	630ms	9s	2s	358ms	135ms		

SonarQube Quality Gate

SimpleCustomerApp Passed

server-side processing: Success

Permalinks

- Last build (#22), 6 min 41 sec ago
- Last stable build (#22), 6 min 41 sec ago

BUILDS >

Filter

Today

✔ #22
8:37 PM
🔍 ▼

✘ #21
8:32 PM
🔍 ▼

✔ #20
8:18 PM
🔍 ▼

✘ #19
8:18 PM
🔍 ▼

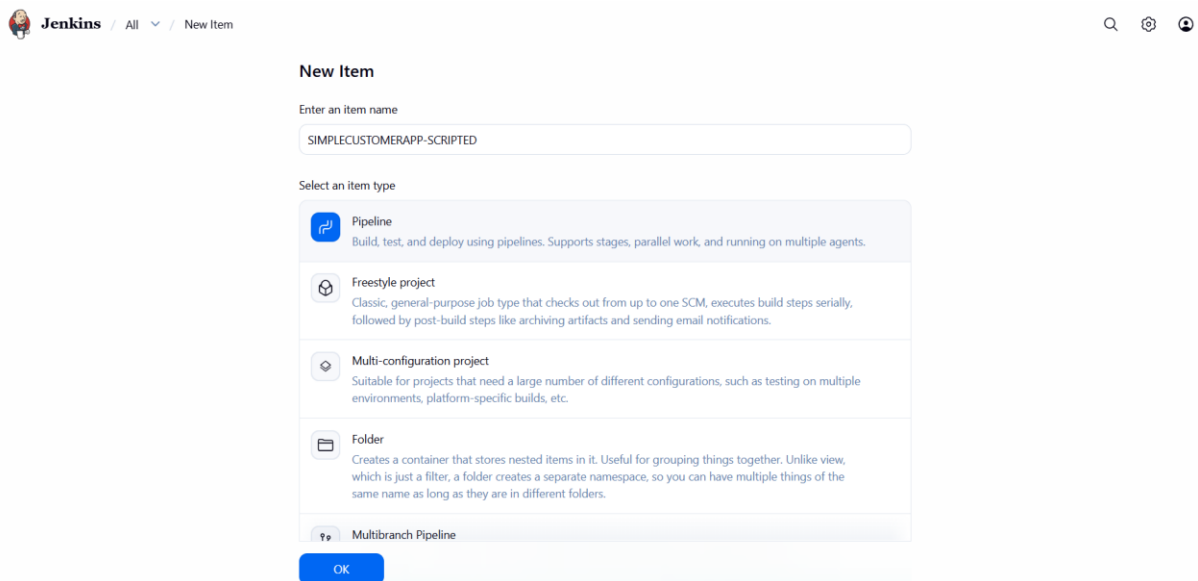
✔ #18
8:06 PM
🔍 ▼

✔ #17
7:52 PM
🔍 ▼

3) Setup a Jenkins CICD pipeline using Scripted pipeline using feature-1.1 branch.

Source Code: https://github.com/betawins/sabear_simplecutomerapp/tree/feature-1.1
stages:

- Git Clone
- SonarQube Integration
- Maven Compilation
- Nexus Artifactory
- Slack Notification
- Deploy On tomcat



- Opened **Configure → Pipeline**
- Used **Scripted Pipeline** syntax
- Defined environment variables:
 - Maven home and PATH
 - Sonar project key and name
 - Nexus URL and repository
 - Tomcat URL
- Added **Slack notification – Start**
- Added **Git Clone** stage with GitHub repository URL

JenkinsSIMPLECUSTOMERAPP-S...Configure

Configure

General

Triggers

Pipeline

Advanced

☐ Build after other projects are built ?

☐ Build periodically ?

☐ GitHub hook trigger for GITSCm polling ?

☐ Poll SCM ?

☐ Trigger builds remotely (e.g., from scripts) ?

Pipeline

Define your Pipeline using Groovy directly or pull it from source control.

Definition

Pipeline script

Script ?

1~ node('master') {
2
3~ stage('Git Clone') {
4 git branch: 'feature-1.1',
5 url: 'https://github.com/betawins/sabear_simplecustomerapp.git'
6 }
7 }
8

try sample Pipeline...

JenkinsSIMPLECUSTOMERAPP-S...

Status

Changes

Build Now

Delete Pipeline

Configure

Rename

Full Stage View

Stages

Pipeline Syntax

Builds >

Filter

Today

#1 7:27 AM

SIMPLECUSTOMERAPP-SCRIPTED

Stage View

Average stage times:
(full run time: ~1s)

Git Clone
622ms

622ms

Permalinks

- Last build (#1). 1 min 20 sec ago
- Last stable build (#1). 1 min 20 sec ago
- Last successful build (#1). 1 min 20 sec ago
- Last completed build (#1). 1 min 20 sec ago

JenkinsSIMPLECUSTOMERAPP-S...#1

Console

Changes

Console Output

Delete build '#1'

Git Build Data

Pipeline Overview

Edit build information

Replay

Pipeline Steps

Workspaces

Started by user BOORA SRIHARI

[Pipeline] Start of Pipeline

[Pipeline] node

Running on Jenkins in /var/lib/jenkins/workspace/SIMPLECUSTOMERAPP-SCRIPTED

[Pipeline] {

[Pipeline] stage

[Pipeline] { (Git Clone)

[Pipeline] git

The recommended git tool is: NONE

No credentials specified

Cloning the remote Git repository

Cloning repository https://github.com/betawins/sabear_simplecustomerapp.git

> git init /var/lib/jenkins/workspace/SIMPLECUSTOMERAPP-SCRIPTED # timeout=10

Fetching upstream changes from https://github.com/betawins/sabear_simplecustomerapp.git

> git --version # timeout=10

> git --version # 'git version 2.50.1'

> git fetch --tags --force --progress -- https://github.com/betawins/sabear_simplecustomerapp.git +refs/heads/*:refs/remotes/origin/* # timeout=10

> git config remote.origin.url https://github.com/betawins/sabear_simplecustomerapp.git # timeout=10

> git config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeout=10

Avoid second fetch

> git rev-parse refs/remotes/origin/feature-1.1^{commit} # timeout=10

Checking out Revision 05222fef4b409f48dd73a8a1b6fbb6c619d398a (refs/remotes/origin/feature-1.1)

> git config core.sparsecheckout # timeout=10

> git checkout -f 05222fef4b409f48dd73a8a1b6fbb6c619d398a # timeout=10

> git branch -a -v --no-abbrev # timeout=10

> git checkout -b feature-1.1 05222fef4b409f48dd73a8a1b6fbb6c619d398a # timeout=10

Commit message: "Update pom.xml"

First time build. Skipping changelog.

[Pipeline] }

[Pipeline] // stage

[Pipeline] }

[Pipeline] // node

[Pipeline] End of Pipeline

Finished: SUCCESS

Jenkins

SIMPLECUSTOMERAPP-S...

Configure

Configure

General

Triggers

Pipeline

Advanced

☐ Trigger builds remotely (e.g., from scripts) ?

Pipeline

Define your Pipeline using Groovy directly or pull it from source control.

Definition

Pipeline script

Script ?

5url: 'https://github.com/betawins/sabear_simplecustomerapp.git'

6}

7

8stage('SonarQube Integration') {

9withSonarQubeEnv('SonarQube') {

10sh '''

11mvn clean verify \

12org.sonarsource.scanner.maven:sonar-maven-plugin:sonar \

13-Dsonar.projectKey=SimpleCustomerApp \

14-Dsonar.projectName=SimpleCustomerApp

15'''

16}

17}

18

19}

try sample Pipeline...

☒ Use Groovy Sandbox ?

Pipeline Syntax

Jenkins

SIMPLECUSTOMERAPP-S...

#5

Starting the text and secrets analysis

1 source file to be analyzed for the text and secrets analysis

1/1 source file has been analyzed for the text and secrets analysis

Sensor TextAndSecretsSensor [text] (done) | time=500ms

Run sensors on project

Sensor IaC Project Sensor [iac]

Sensor IaC Project Sensor [iac] (done) | time=7ms

Sensor Zero Coverage Sensor

Sensor Zero Coverage Sensor (done) | time=0ms

Gather SCA dependencies on project

Dependency analysis skipped

CPD Executor Calculating CPD for 0 files

CPD Executor CPD calculation finished (done) | time=0ms

SCH revision ID '05222fef4b498f40dd73a8a1befbdbc6c19d398a'

Analysis report generated in 139ms, dir size=289.5 kB

Analysis report compressed in 15ms, zip size=30.8 kB

Analysis report uploaded in 63ms

ANALYSIS SUCCESSFUL, you can find the results at: <http://44.197.128.43:9000/dashboard?tid=SimpleCustomerApp>

Note that you will be able to access the updated dashboard once the server has processed the submitted analysis report

More about the report processing at <http://44.197.128.43:9000/api/ce/task?id=062fda12-d12b-4965-a645-f18021168800>

Analysis total time: 3.938 s

SonarScanner Engine completed successfully

BUILD SUCCESS

Total time: 10.264 s

Finished at: 2026-08-19T08:11:48Z

Pipeline }

Pipeline // withSonarQubeEnv

Pipeline }

Pipeline // stage

Pipeline }

Pipeline // node

Pipeline End of Pipeline

Finished: SUCCESS

Jenkins

SIMPLECUSTOMERAPP-S...

Stage View

Status

Changes

Build Now

Delete Pipeline

Configure

Rename

Full Stage View

SonarQube

Stages

Pipeline Syntax

Builds >

Filter

Today

#58:11 AM

#48:04 AM

#37:52 AM

#27:31 AM

#17:27 AM

✓ SIMPLECUSTOMERAPP-SCRIPTED

Average stage times:
(full run time: ~7s)

	Git Clone	SonarQube Integration
#5Aug 19 13:41No Changes	460ms	12s
#4Aug 19 13:34No Changes	251ms	5s failed
#3Aug 19 13:22No Changes	253ms	20s failed
#2Aug 19 13:01No Changes	517ms	21s failed
#1Aug 19 12:57No Changes	622ms	

Jenkins

SIMPLECUSTOMERAPP-S...

Configure

Configure

General

Triggers

Pipeline

Advanced

Pipeline

Define your Pipeline using Groovy directly or pull it from source control.

Definition

Pipeline script

Script ?

```
9~ withSonarQubeEnv('SonarQube') {
10   ...
11   sh
12     mvn clean verify \
13       org.sonarsource.scanner.maven:sonar-maven-plugin:sonar \
14       -Dsonar.projectKey=SimpleCustomerApp \
15       -Dsonar.projectName=SimpleCustomerApp
16   ...
17 }
18
19~ stage('Haven Compilation') {
20   sh 'mvn clean package -DskipTests'
21 }
22
23 }
```

try sample Pipeline...

☒ Use Groovy Sandbox ?

Pipeline Syntax

Jenkins

SIMPLECUSTOMERAPP-S...

#6

```
[INFO] Deleting /var/lib/jenkins/workspace/SIMPLECUSTOMERAPP-SCRIPTED/target
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ SimpleCustomerApp ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /var/lib/jenkins/workspace/SIMPLECUSTOMERAPP-SCRIPTED/src/main/resources
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ SimpleCustomerApp ---
[INFO] No sources to compile
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ SimpleCustomerApp ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /var/lib/jenkins/workspace/SIMPLECUSTOMERAPP-SCRIPTED/src/test/resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ SimpleCustomerApp ---
[INFO] No sources to compile
[INFO]
[INFO] --- surefire:3.5.4:test (default-test) @ SimpleCustomerApp ---
[INFO] Tests are skipped.
[INFO]
[INFO] --- war:3.3.2:war (default-war) @ SimpleCustomerApp ---
[INFO] Packaging webapp
[INFO] Assembling webapp [SimpleCustomerApp] in [/var/lib/jenkins/workspace/SIMPLECUSTOMERAPP-SCRIPTED/target/SimpleCustomerApp-6-SNAPSHOT]
[INFO] Processing war project
[INFO] Building war: /var/lib/jenkins/workspace/SIMPLECUSTOMERAPP-SCRIPTED/target/SimpleCustomerApp-6-SNAPSHOT.war
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 1.427 s
[INFO] Finished at: 2026-08-19T09:14:09Z
[INFO]
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Jenkins

SIMPLECUSTOMERAPP-S...

Status

Changes

Build Now

Delete Pipeline

Configure

Rename

Full Stage View

SonarQube

Stages

Pipeline Syntax

Builds >

Filter

Today

#6

9:13 AM

#5

8:11 AM

Stage View

Average stage times:
(full run time: ~9s)

	Git Clone	SonarQube Integration	Maven Compilation
	434ms	13s	2s
#6 Aug 19 14:43 No Changes	504ms	10s	2s
#5 Aug 19 13:41 No Changes	460ms	12s	
#4 Aug 19 13:34 No Changes	251ms	5s failed	
#3 Aug 19 13:22 No Changes	253ms	20s failed	

Configure

- General
- Triggers
- Pipeline
- Advanced

Define your Pipeline using Groovy directly or pull it from source control.

Definition

Pipeline script

```
23 stage('Nexus Artifactory') {
24   withCredentials([
25     usernamePassword(
26       credentialsId: 'nexus-credentials',
27       usernameVariable: 'NEXUS_USERNAME',
28       passwordVariable: 'NEXUS_PASSWORD'
29     )
30   ]) {
31     sh '''
32       WAR_FILE=$(ls target/*.war)
33
34       curl -u "$NEXUS_USERNAME:$NEXUS_PASSWORD" \
35         --upload-file "$WAR_FILE" \
36         "http://44.197.120.43:8081/repository/simpleCustomerApp/${basename "$WAR_FILE"}"
37     '''
38   }
39 }
```

☒ Use Groovy Sandbox

[Pipeline Syntax](#)

Advanced

Advanced

Status

- <> Changes
- ▶ Build Now
- 🗑 Delete Pipeline
- ⚙ Configure
- ✎ Rename
- 🔍 Full Stage View
- 🔍 SonarQube
- 📊 Stages
- 🔗 Pipeline Syntax

Builds

Filter

Today

#8 9:30 AM

#7 9:25 AM

#6 9:13 AM

SIMPLECUSTOMERAPP-SCRIPTED

Stage View

	Git Clone	SonarQube Integration	Maven Compilation	Nexus Artifactory
Average stage times: (full run time: ~10s)	429ms	12s	2s	384ms
#8 Aug 19 15:00 No Changes	220ms	9s	2s	350ms
#7 Aug 19 14:55 No Changes	605ms	10s	2s	419ms failed
#6 Aug 19 14:43 No Changes	504ms	10s	2s	
#5 Aug 19 13:41 No Changes	460ms	12s		
#4 Aug 19 No	251ms	5s		

```
[INFO] --- war:3.3.2:war (default-war) @ SimpleCustomerApp ---
[INFO] Packaging webapp
[INFO] Assembling webapp [SimpleCustomerApp] in [/var/lib/jenkins/workspace/SIMPLECUSTOMERAPP-SCRIPTED/target/SimpleCustomerApp-8-SNAPSHOT]
[INFO] Processing war project
[INFO] Building war: /var/lib/jenkins/workspace/SIMPLECUSTOMERAPP-SCRIPTED/target/SimpleCustomerApp-8-SNAPSHOT.war
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 1.390 s
[INFO] Finished at: 2026-08-19T09:30:43Z
[INFO] -----
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Nexus Artifactory)
[Pipeline] withCredentials
Masking supported pattern matches of $NEXUS_PASSWORD
[Pipeline] {
[Pipeline] sh
++ ls target/SimpleCustomerApp-8-SNAPSHOT.war
++ WAR_FILE=target/SimpleCustomerApp-8-SNAPSHOT.war
++ basename target/SimpleCustomerApp-8-SNAPSHOT.war
++ curl -u admin:**** --upload-file target/SimpleCustomerApp-8-SNAPSHOT.war http://32.195.55.5:8081/repository/simpleCustomerApp/SimpleCustomerApp-8-SNAPSHOT.war
% Total    % Received % Xferd Average Speed   Time    Time     Current
           Dload  Upload   Total   Spent    Left  Speed

  0    0    0    0    0    0     0  0  0  0  0  0 --:--:-- --:--:-- --:--:--    0
100 1421    0    0 100 1421    0 9878 --:--:-- --:--:-- --:--:-- 9937
[Pipeline] }
[Pipeline] // withCredentials
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Configure

- General
- Triggers
- Pipeline
- Advanced

Script ?

```
34 curl -u "$NEXUS_USERNAME:$NEXUS_PASSWORD" \
35   --upload-file "$WAR_FILE" \
36   "http://44.197.128.43:8081/repository/simpleCustomerApp/${basename "$WAR_FILE"}"
37   ...
38 }
39 }
40
41 stage('Slack Notification') {
42   slackSend(
43     channel: '#build_notification',
44     message: "SIMPLECUSTOMERAPP-SCRIPTED build #${env.BUILD_NUMBER} completed successfully.",
45     tokenCredentialId: 'slack_notification_token'
46   )
47 }
48 }
```

☒ Use Groovy Sandbox ?

[Pipeline Syntax](#)

Advanced

Advanced ▾

Save

Apply

Status

- Changes
- Build Now
- Delete Pipeline
- Configure
- Rename
- Full Stage View
- SonarQube
- Stages
- Pipeline Syntax

Builds ▾

Filter

Today

- #10 9:42 AM
- #9 9:40 AM
- #8 9:30 AM
- #7 9:25 AM

✓ SIMPLECUSTOMERAPP-SCRIPTED

Stage View

		Git Clone	SonarQube Integration	Maven Compilation	Nexus Artifactory	Slack Notification
	Average stage times: (full run time: ~11s)	386ms	12s	2s	370ms	433ms
#10	Aug 19 15:12 No Changes	214ms	10s	Success	354ms	433ms
#9	Aug 19 15:10 No Changes	217ms	9s	2s	360ms failed	
#8	Aug 19 15:00 No Changes	220ms	9s	2s	350ms	
#7	Aug 19 14:55 No Changes	605ms	10s	2s	419ms failed	
#6	Aug 19 14:43 No Changes	504ms	10s	2s		

```
[INFO] -----
[INFO] Total time: 1.398 s
[INFO] Finished at: 2026-08-19T09:42:39Z
[INFO] -----
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Nexus Artifactory)
[Pipeline] withCredentials
Masking supported pattern matches of $NEXUS_PASSWORD
[Pipeline] {
[Pipeline] sh
++ ls target/SimpleCustomerApp-10-SNAPSHOT.war
+ WAR_FILE=target/SimpleCustomerApp-10-SNAPSHOT.war
++ basename target/SimpleCustomerApp-10-SNAPSHOT.war
+ curl -u admin:**** --upload-file target/SimpleCustomerApp-10-SNAPSHOT.war http://32.195.55.5:8081/repository/simpleCustomerApp/SimpleCustomerApp-10-SNAPSHOT.war
% Total    % Received % Xferd Average Speed   Time    Time     Current
                                 Dload  Upload   Total   Spent    Left   Speed

  0   0    0    0    0     0      0      0  0  0  0  0  0  0  0  0  0  0  0
100 1422  0    0 100 1422    0 39434   --:--:-- --:--:-- --:--:-- 39500
[Pipeline] }
[Pipeline] // withCredentials
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Slack Notification)
[Pipeline] slackSend
Slack Send Pipeline step running, values are - baseUrl: <empty>, teamDomain: harl, channel: #build_notification, color: <empty>, botUser: true, tokenCredentialId: slack_notif
notifyCommiters: false, iconEmoji: <empty>, username: <empty>, timestamp: <empty>
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

H

Hari

An offer awaits

Find a conversation...

Huddles

Directories

Starred

Drag and drop important stuff here

Channels

all-hari

build_notification

new-channel

social

+ Add channels

Direct messages

srihari48326 you

+ Invite people

Agents & apps

Jenkins

Slack

build_notification

Invite teammates Huddle

Messages Add canvas

BETAWINS-HIRING-APP - #11 Success after 15 sec (Open) + 1 new message

Today

Jenkins APP 1:47 AM

Slack/Jenkins plugin: you're all set on <http://44.199.226.166:8081/>

FAILED: SIMPLECUSTOMERAPP #19 - Pipeline failed. Check Jenkins Console Output.

SUCCESS: SIMPLECUSTOMERAPP #20 - Build and Nexus upload completed successfully.

Jenkins APP 2:03 AM

FAILED: SIMPLECUSTOMERAPP #21 - Pipeline failed. Check Console Output.

SUCCESS: SIMPLECUSTOMERAPP #22 - Build and Nexus upload completed.

SUCCESS: SIMPLECUSTOMERAPP #22 - Application deployed to Tomcat successfully.

Jenkins APP 3:12 PM

SIMPLECUSTOMERAPP-SCRIPTED build #10 completed successfully.

B I U

Message #build_notification

+

Jenkins

SIMPLECUSTOMERAPP-S...

Configure

Q

Configure

General

Triggers

Pipeline

Advanced

Script

```
54 stage('Deploy On Tomcat') {
55   withCredentials([
56     usernamePassword(
57       credentialsId: 'tomcat-credentials',
58       usernameVariable: 'TOPCAT_USERNAME',
59       passwordVariable: 'TOPCAT_PASSWORD'
60     )
61   ]) {
62     sh '''
63       WAR_FILE=$(ls target/*.war)
64
65       echo "Deploying $WAR_FILE to Tomcat..."
66
67       curl --fail \
68         -u "$TOPCAT_USERNAME:$TOPCAT_PASSWORD" \
```

try sample Pipeline...

☒ Use Groovy Sandbox

Pipeline Syntax

Advanced

Advanced

Save Apply

Not secure 44.203.25.236:8081/job/SIMPLECUSTOMERAPP-SCRIPTED/

Jenkins SIMPLECUSTOMERAPP-S...

STATUS SIMPLECUSTOMERAPP-SCRIPTED

Changes
Build Now
Delete Pipeline
Configure
Rename
Full Stage View
SonarQube
Stages
Pipeline Syntax

BUILDS

Filter

Today

- #12 9:56 AM
- #11 9:53 AM
- #10 9:42 AM
- #9 9:40 AM

Stage View

	Git Clone	SonarQube Integration	Maven Compilation	Nexus Artifactory	Slack Notification	Deploy On Tomcat
Average stage times: (Full run time: ~13s)	316ms	10s	2s	363ms	232ms	205ms
#12 Aug 19 15:26 No Changes	213ms	10s	2s	344ms	143ms	350ms
#11 Aug 19 15:23 No Changes	229ms	9s	2s	351ms	122ms	61ms
#10 Aug 19 15:12 No Changes	214ms	10s	2s	354ms	433ms	
#9 Aug 19 15:10 No Changes	217ms	9s	2s	360ms		
#8 Aug 19 15:00 No Changes	220ms	9s	2s	350ms		

```
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Slack Notification)
[Pipeline] slackSend
Slack Send Pipeline step running, values are - baseUrl: <empty>, teamDomain: hari, channel: #build_notification, color: <empty>, botUser: true, tokenCredentialId: slack_notification_token, notifyCommiters: false, iconEmoji: <empty>, username: <empty>, timestamp: <empty>
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy On Tomcat)
[Pipeline] withCredentials
Masking supported pattern matches of $TOMCAT_PASSWORD
[Pipeline] {
[Pipeline] sh
++ ls target/SimpleCustomerApp-12-SNAPSHOT.war
++ WAR_FILE=target/SimpleCustomerApp-12-SNAPSHOT.war
+ echo 'Deploying target/SimpleCustomerApp-12-SNAPSHOT.war to Tomcat...'
Deploying target/SimpleCustomerApp-12-SNAPSHOT.war to Tomcat...
+ curl --fail -u ****:* --upload-file target/SimpleCustomerApp-12-SNAPSHOT.war 'http://32.195.55.5:8080/manager/text/deploy?path=/SimpleCustomerApp&update=true'
% Total    % Received % Xferd Average Speed   Time    Time     Time
             Dload  Upload   Total   Spent    Left  Speed

  0     0    0     0    0     0      0      0      0  0:00:00  0:00:00  0:00:00
100 1485    0    63 100 1422 1862 42832    0:00:00  0:00:00  0:00:00 45800
OK - Deployed application at context path [/SimpleCustomerApp]
+ echo 'Tomcat deployment completed successfully.'
Tomcat deployment completed successfully.
[Pipeline] }
[Pipeline] // withCredentials
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

- Logged into Jenkins server via SSH
- Navigated to:
 - /var/lib/jenkins/workspace
- Verified directories:
 - SIMPLECUSTOMERAPP
 - scripted_pipeline
- Confirms Jenkins job workspace creation



Scripted Pipeline (Jenkinsfile - Skeleton)

```
Groovy
node {
    stage('Checkout') {
        echo 'Cloning source code'
        // git 'https://github.com/org/repo.git'
    }

    stage('Build') {
        echo 'Building the application'
        // sh 'mvn clean package'
    }

    stage('Test') {
        echo 'Running tests'
        // sh 'mvn test'
    }

    stage('Quality Analysis') {
        echo 'SonarQube analysis'
        // sh 'sonar-scanner'
    }

    stage('Package') {
        echo 'Packaging artifact'
    }

    stage('Deploy') {
        echo 'Deploying application'
    }
}
```

```

35 Declarative Pipeline (Jenkinsfile - Skeleton)
36
37 pipeline {
38     agent any
39
40     environment {
41         // Example:
42         // JAVA_HOME = '/usr/lib/jvm/java-17'
43     }
44
45     stages {
46         stage('Checkout') {
47             steps {
48                 echo 'Cloning source code'
49                 // git 'https://github.com/org/repo.git'
50             }
51         }
52
53         stage('Build') {
54             steps {
55                 echo 'Building application'
56                 // sh 'mvn clean package'
57             }
58         }
59
60         stage('Test') {
61             steps {
62                 echo 'Running tests'
63                 // sh 'mvn test'
64             }
65         }
66
67         stage('Quality Analysis') {
68             steps {
69                 echo 'Running SonarQube analysis'
70             }
71         }
72
73         stage('Deploy') {
74             steps {
75                 echo 'Deploying application'
76             }
77         }
78     }
79
80     post {
81         success {
82             echo 'Pipeline completed successfully'
83         }
84         failure {
85             echo 'Pipeline failed'
86         }
87     }
88 }

```

5) Create a parameterized job in Jenkins.

Source Code: <https://github.com/betawins/spring3-mvc-maven-xml-hello-world-1.git>

- Opened **Jenkins** → **New Item**
- Entered job name: **spring3-parameterized-job**
- Selected **Freestyle project**
- Clicked **OK**

The image shows two screenshots from the Jenkins web interface. The top screenshot is the 'New Item' page. It has a header 'Jenkins / All / New Item'. The main section is titled 'New Item' and contains a form to 'Enter an item name' with the value 'spring3-parameterized-job'. Below this is a 'Select an item type' section with several options: Pipeline, Freestyle project (selected), Multi-configuration project, Folder, Multibranch Pipeline, and Organization Folder. At the bottom is a 'Duplicate an existing item' button and an 'OK' button. The bottom screenshot is the 'Configure' page for the 'spring3-parameterized-job'. It has a header 'Jenkins / spring3-parameterized-job / Configure'. The left sidebar shows 'Configure' with sub-items: General (selected), Source Code Management, Triggers, Environment, Build Steps, and Post-build Actions. The main area shows configuration options: 'Enable project-based security' (unchecked), 'Discard old builds' (unchecked), 'GitHub project' (unchecked), and 'This project is parameterized' (checked). Below the checked option is a '+ Add Parameter' button and a dropdown menu showing various parameter types: Filter, Boolean Parameter, Choice Parameter (selected), Credentials Parameter, File Parameter, Multi-line String Parameter, Password Parameter, Run Parameter, and String Parameter. At the bottom, there is a 'Source Code Management' section with a description: 'Connect and manage your code repository to automatically pull the latest code for your builds.'

- Opened **Configure** → **General**
- Enabled **This project is parameterized**
- Added **Choice Parameter**
- Parameter name: **ENV**
- Choices added:
 - dev
 - qa
 - prod

- Saved parameter configuration

The screenshot shows the Jenkins 'Configure' page for a job named 'spring3-parameterized-job'. The left sidebar contains a 'Configure' section with a list of tabs: General, Source Code Management, Triggers, Environment, Build Steps, and Post-build Actions. The 'General' tab is selected. In the main area, there are several checkboxes: 'Enable project-based security' (unchecked), 'Discard old builds' (unchecked), and 'GitHub project' (unchecked). Below these, the checkbox 'This project is parameterized' is checked. A 'Choice Parameter' configuration box is visible, containing a 'Name' field with the value 'ENV', a 'Choices' list with 'dev' and 'prod', and a 'Description' field. At the bottom of the configuration area, there is a '+ Add Parameter' button. At the very bottom of the page, there are 'Save' and 'Apply' buttons.

- Added **String Parameter**
- Parameter name: GIT_BRANCH
- Default value: master
- Description: *Git branch to build*
- Used to dynamically select Git branch

The first screenshot shows the 'Configure' page for the 'spring3-parameterized-job' in Jenkins. The left sidebar has tabs for 'General', 'Source Code Management', 'Triggers', 'Environment', 'Build Steps', and 'Post-build Actions'. The 'String Parameter' section is active, showing a parameter named 'GIT_BRANCH' with a default value of 'master' and a description 'git branch to build'. There is a 'Trim the string' checkbox and an 'Add Parameter' button at the bottom.

The second screenshot shows the 'Source Code Management' tab. It has a heading 'Connect and manage your code repository to automatically pull the latest code for your builds.' with radio buttons for 'None' and 'Git'. The 'Git' option is selected. Below, there's a 'Repositories' section with a 'Repository URL' field containing 'https://github.com/betawins/spring3-mvc-maven-xml-hello-world-1.git', a 'Credentials' dropdown set to 'none', and an 'Add' button. An 'Advanced' dropdown is also present. Below the repositories is an 'Add Repository' button. The 'Branches to build' section has a 'Branch Specifier (blank for \'any\')' field containing '*/\${GIT_BRANCH}'. There is an 'Add Branch' button. At the bottom are 'Save' and 'Apply' buttons.

- Opened **Build Steps**
- Added **Invoke top-level Maven targets**
- Selected Maven installation: MVN_HOME
- Maven goals configured:
 - clean package -DskipTests=\${SKIP_TESTS}
- Build behavior controlled via parameters
- Opened **Post-build Actions**
- Added **Slack Notifications**
- Enabled:
 - Notify Build Start
 - Notify Success
- Linked Jenkins job notifications to Slack

Jenkins / spring3-parameterized-job / Configure

Configure

- General
- Source Code Management
- Triggers
- Environment**
- Build Steps
- Post-build Actions

☐ Delete workspace before build starts

☐ Use secret text(s) or file(s) ?

☐ Add timestamps to the Console Output

☐ Inspect build log for published build scans

☐ Terminate a build if it's stuck

☒ With Ant ?

JDK

java8

Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment.

Invoke top-level Maven targets ?

Maven Version

MVN_HOME

Goals

clean package -DskipTests=\${SKIP_TEST}

Advanced

+ Add build step

Save Apply

- Clicked **Build with Parameters**
- Selected:
- ENV = dev
- GIT_BRANCH = master
- SKIP_TESTS = true
- Clicked **Build**

Jenkins

spring3-parameterized-job

Status

Changes

Workspace

Build with Parameters

Delete Project

Configure

Rename

Builds

No builds

Project spring3-parameterized-job

This build requires parameters:

ENV

dev

GIT_BRANCH

Git branch to build

master

☒ SKIP_TESTS

Skip unit tests

Build

Cancel

Jenkins

spring3-parameterized-job

Status

Changes

Workspace

Build with Parameters

Delete Project

Configure

Rename

Builds

Filter

Today

#7 2:49 PM

#6 2:36 PM

#5 2:24 PM

#4 2:19 PM

#3 2:17 PM

#2 2:09 PM

#1 2:04 PM

spring3-parameterized-job

Permalinks

Last build (#7), 3 min 2 sec ago

Last stable build (#7), 3 min 2 sec ago

Last successful build (#7), 3 min 2 sec ago

Last failed build (#6), 16 min ago

Last unsuccessful build (#6), 16 min ago

Last completed build (#7), 3 min 2 sec ago

Jenkins

spring3-parameterized-job

#7

Progress (1): 7.1/7.4 MB

Progress (1): 7.1/7.4 MB

Progress (1): 7.1/7.4 MB

Progress (1): 7.1/7.4 MB

Progress (1): 7.2/7.4 MB

Progress (1): 7.2/7.4 MB

Progress (1): 7.2/7.4 MB

Progress (1): 7.2/7.4 MB

Progress (1): 7.2/7.4 MB

Progress (1): 7.2/7.4 MB

Progress (1): 7.3/7.4 MB

Progress (1): 7.3/7.4 MB

Progress (1): 7.3/7.4 MB

Progress (1): 7.3/7.4 MB

Progress (1): 7.4/7.4 MB

Progress (1): 7.4/7.4 MB

Progress (1): 7.4/7.4 MB

Progress (1): 7.4 MB

Downloaded from central: <https://repo.maven.apache.org/maven2/com/github/luben/zstd-jni/1.5.7-6/zstd-jni-1.5.7-6.jar> (7.4 MB at 19 MB/s)

[INFO] Packaging webapp

[INFO] Assembling webapp [ncodeit-hello-world] in [/var/lib/jenkins/workspace/spring3-parameterized-job/target/ncodeit-hello-world-3.0]

[INFO] Processing war project

[INFO] Copying webapp resources [/var/lib/jenkins/workspace/spring3-parameterized-job/src/main/webapp]

[INFO] Building war: /var/lib/jenkins/workspace/spring3-parameterized-job/target/ncodeit-hello-world-3.0.war

[INFO] -----

[INFO] BUILD SUCCESS

[INFO] -----

[INFO] Total time: 4.676 s

[INFO] Finished at: 2026-08-19T14:49:41Z

[INFO] -----

[Slack Notifications] found #6 as previous completed, non-aborted build

[Slack Notifications] will send OnSuccessNotification because build matches and user preferences allow it

Finished: SUCCESS

Files

Channels

all-hari

build_notification

new-channel

social

+ Add channels

Direct messages

spring3-parameterized-job - #5 Still Failing after 5.7 sec (Open)

Jenkins APP 8:06 PM

spring3-parameterized-job - #6 Started by user BOORA SRIHARI (Open)

spring3-parameterized-job - #6 Still Failing after 2.9 sec (Open)

Jenkins APP 8:19 PM

spring3-parameterized-job - #7 Started by user BOORA SRIHARI (Open)

8:19

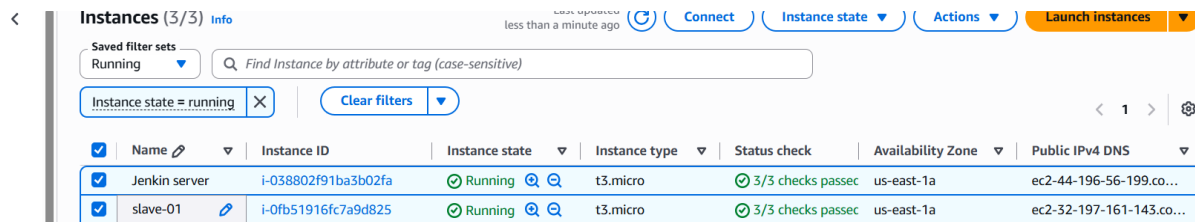
spring3-parameterized-job - #7 Success after 6.3 sec (Open)

6) Setup one slave machine for Jenkins.

EC2 & Initial Setup

1. Launched EC2 instances:

- slave-01



Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS
Jenkin server	i-038802f91ba3b02fa	Running	t3.micro	3/3 checks passed	us-east-1a	ec2-44-196-56-199.co...
slave-01	i-0fb51916fc7a9d825	Running	t3.micro	3/3 checks passed	us-east-1a	ec2-32-197-161-143.co...

Slave-01 OS Configuration

Logged in to slave-01 as root.

Installed Java:

```
sudo yum install java-21-amazon-corretto -y
```

Verified Java installation completed.

```
[ec2-user@ip-172-31-7-219 ~]$ sudo dnf install java-21-amazon-corretto -y
Amazon Linux 2023 Kernel Livepatch repository
Dependencies resolved.
```

Package	Architecture	Version	Repository
Installing:			
java-21-amazon-corretto	x86_64	1:21.0.12+8-1.amzn2023.1	amazonlinux
Installing dependencies:			
alsa-lib	x86_64	1:2.7.2-1.amzn2023.0.3	amazonlinux
cairo	x86_64	1.18.0-4.amzn2023.0.3	amazonlinux
dejavu-sans-fonts	noarch	2.37-16.amzn2023.0.2	amazonlinux
dejavu-sans-mono-fonts	noarch	2.37-16.amzn2023.0.2	amazonlinux
dejavu-serif-fonts	noarch	2.37-16.amzn2023.0.2	amazonlinux
fontconfig	x86_64	2.13.94-2.amzn2023.0.2	amazonlinux
fonts-filesystem	noarch	1:2.0.5-12.amzn2023.0.2	amazonlinux
freetype	x86_64	2.13.2-5.amzn2023.0.2	amazonlinux
glib	x86_64	5.2.1-9.amzn2023.0.4	amazonlinux
google-noto-fonts-common	noarch	20240401-1.amzn2023.0.2	amazonlinux
google-noto-sans-vf-fonts	noarch	20240401-1.amzn2023.0.2	amazonlinux
graphite2	x86_64	1.3.14-7.amzn2023.0.3	amazonlinux
harfbuzz	x86_64	7.0.0-2.amzn2023.0.2	amazonlinux
java-21-amazon-corretto-headless	x86_64	1:21.0.12+8-1.amzn2023.1	amazonlinux
javapackages-filesystem	noarch	6.0.0-7.amzn2023.0.6	amazonlinux
langpacks-core-font-en	noarch	3.0-21.amzn2023.0.4	amazonlinux
libICE	x86_64	1.1.1-3.amzn2023.0.1	amazonlinux
libSM	x86_64	1.2.4-3.amzn2023.0.1	amazonlinux
libX11	x86_64	1.8.10-2.amzn2023.0.1	amazonlinux
libX11-common	noarch	1.8.10-2.amzn2023.0.1	amazonlinux

SSH Configuration on Slave-01

Generated SSH key pair:

```
ssh-keygen
```

Navigated to SSH directory:

```
cd ~/.ssh
```

Added public key to authorized keys:

```
cat id_rsa.pub >> authorized_keys
```

Fixed permissions:

```
chmod 700 authorized_keys
```

```
[ec2-user@ip-172-31-7-219 ~]$ ssh-keygen
Generating public/private rsa key pair.
Enter file in which to save the key (/home/ec2-user/.ssh/id_rsa):
Enter passphrase for "/home/ec2-user/.ssh/id_rsa" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/ec2-user/.ssh/id_rsa
Your public key has been saved in /home/ec2-user/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:zRbe7hfz2Po0+sk+JTX3ZR8sPWk/h3OR4MmGM078ag ec2-user@ip-172-31-7-219.ec2.internal
The key's randomart image is:
+---[RSA 3072]-----+
|
| . . . o
| B o . *o
| S X . +o+
| =...=.o=
| .+oo++=
| oo.+o*=
| E. .=BBoB
+---[SHA256]-----+
[ec2-user@ip-172-31-7-219 ~]$ cd ~/.ssh
[ec2-user@ip-172-31-7-219 .ssh]$ ls
authorized_keys  id_rsa  id_rsa.pub
[ec2-user@ip-172-31-7-219 .ssh]$ cat id_rsa.pub >> authorized_keys
[ec2-user@ip-172-31-7-219 .ssh]$ chmod 700 authorized_keys
[ec2-user@ip-172-31-7-219 .ssh]$
```

Jenkins Master SSH Trust

Logged in to jenkins-master.

Added slave host key:

```
ssh-keyscan -H 13.61.10.105 >> /var/lib/jenkins/.ssh/known_hosts
```

Set correct permissions:

```
chown jenkins:jenkins /var/lib/jenkins/.ssh/known_hosts
```

```
chmod 700 /var/lib/jenkins/.ssh
```

```
chmod 600 /var/lib/jenkins/.ssh/known_hosts
```

```
[root@ip-172-31-4-173 ec2-user]# mkdir -p /var/lib/jenkins/.ssh
[root@ip-172-31-4-173 ec2-user]# cd /var/lib/jenkins/.ssh
bash: cd: /var/lib/jenkins/.ssh: No such file or directory
[root@ip-172-31-4-173 ec2-user]# cd /var/lib/jenkins/.ssh
[root@ip-172-31-4-173 .ssh]# ssh-keyscan -H 32.197.161.143 >> /var/lib/jenkins/.ssh/known_hosts
[root@ip-172-31-4-173 .ssh]# chown jenkins:jenkins known_hosts
[root@ip-172-31-4-173 .ssh]# chmod 700 known_hosts
[root@ip-172-31-4-173 .ssh]#
```

Jenkins Node Creation (slave-01)

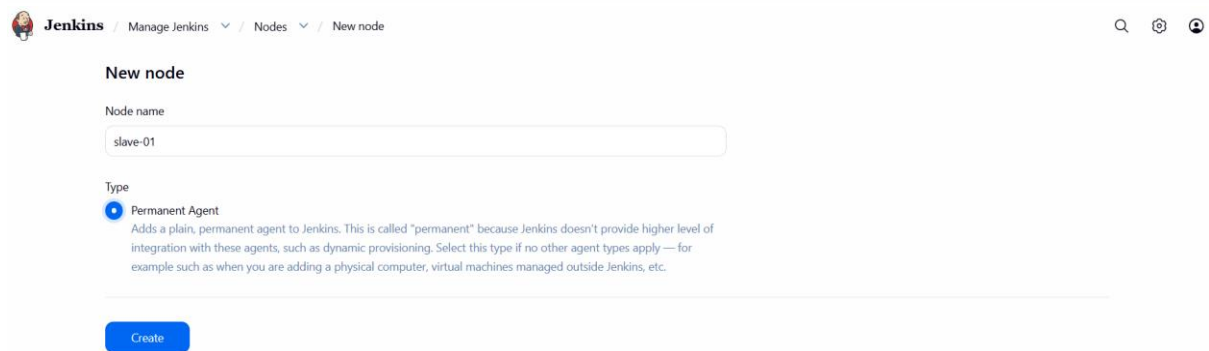
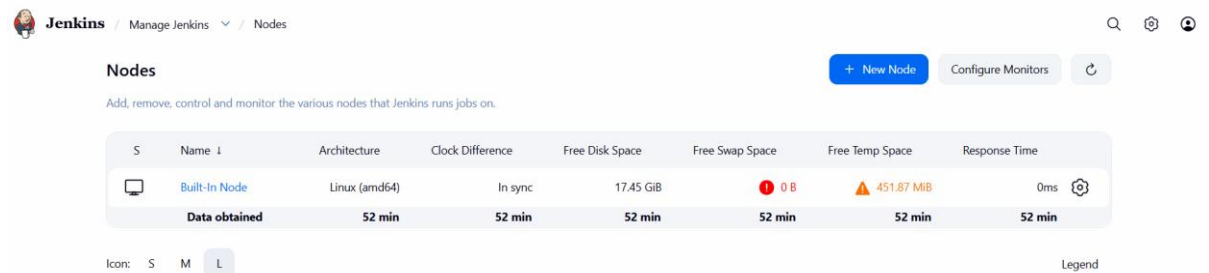
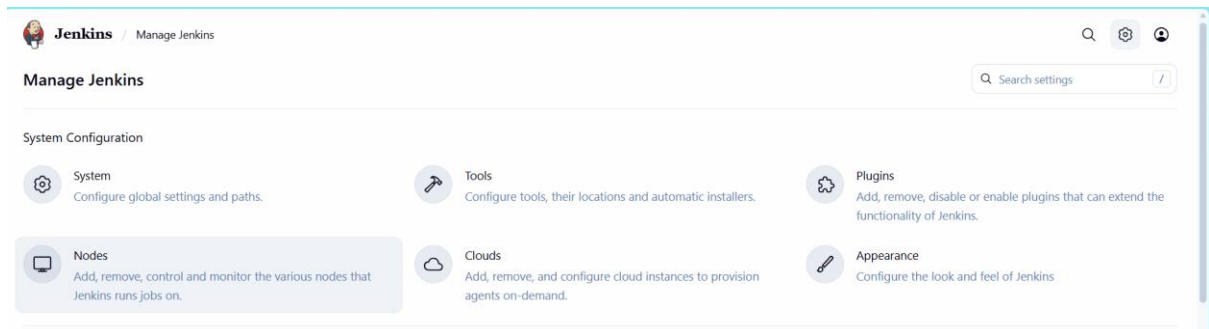
Opened Jenkins dashboard.

Navigated to **Manage Jenkins** → **Nodes**.

Clicked **New Node**.

Entered node name: slave-01.

Selected **Permanent Agent** and created the node.



Node Configuration (slave-01)

Set **Number of executors** to 2.

Initially observed agent offline due to root directory permission issue.

Logged in to slave-01 and created Jenkins workspace:

```
mkdir -p /home/ec2-user/jenkins
```

```
chown -R ec2-user:ec2-user /home/ec2-user/jenkins
```

Updated **Remote root directory** in Jenkins to:

```
/home/ec2-user/jenkins
```

Added **Label**:

```
java
```

Set **Usage** to:

Only build jobs with label expressions matching this node

 **Jenkins** / Manage Jenkins / Nodes

Name 
slave-01

Description 

Plain text [Preview](#)

Number of executors 
2

Remote root directory 
/root

Labels 
java

Usage 
Only build jobs with label expressions matching this node

Launch method 
Launch agents via SSH

Host 
32.197.161.143

Credentials 
ec2-user  

Host Key Verification Strategy 
Known hosts file Verification Strategy  



Availability 
Keep this agent online as much as possible



SSH Launch Configuration

Selected **Launch agents via SSH**.

Entered **Host**: 13.61.10.105.

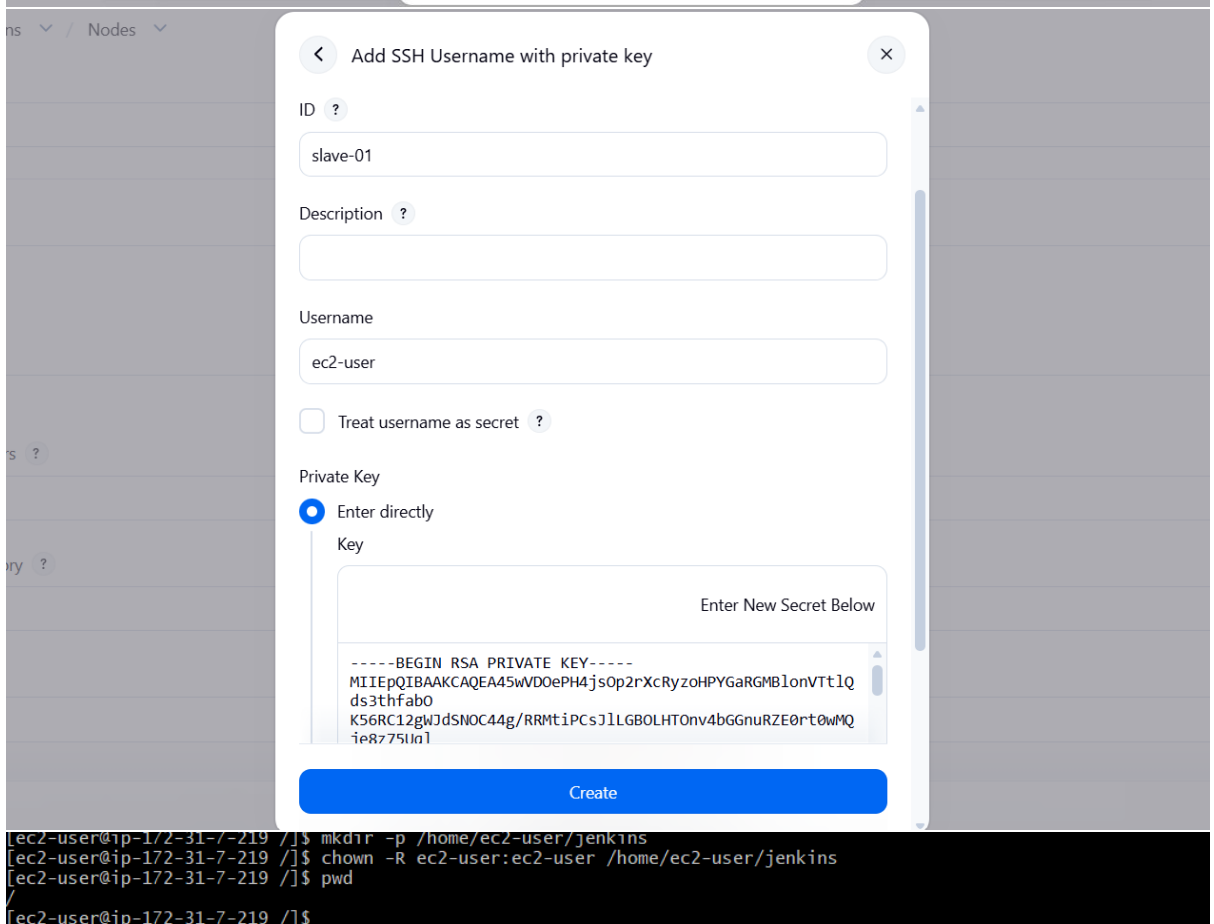
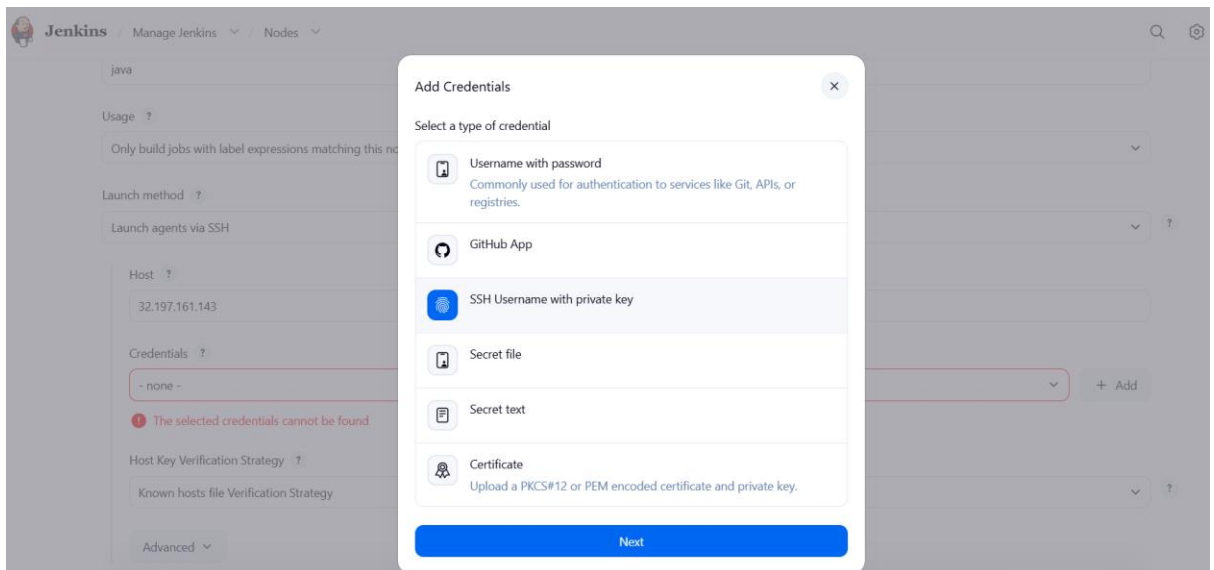
Selected SSH **credentials (ec2-user)**.

Chose **Known hosts file Verification Strategy**.

Saved configuration.

Verified SSH authentication successful in node logs.

Confirmed slave-01 status changed to **online**.



Status

Delete Agent

Configure

Build History

Load Statistics

Log

Build Executor Status

0/2

slave-01

offline

Warning: no key algorithms provided; JENKINS-42959 disabled

SSHLauncher{host='32.197.161.143', port=22, credentialsId='slave-01', jvmOptions='', javaPath='', prefixStartsSlaveCmd='', suffixStartsSlaveCmd='', launchTimeoutSeconds=60, maxNumRetries=10, retryWaitTime=15, sshHostKeyVerificationStrategy=hudson.plugins.sshslaves.verifiers.KnownHostsFileKeyVerificationStrategy, tcpNoDelay=true, trackCredentials=true}

[08/11/26 14:23:22] [SSH] Opening SSH connection to 32.197.161.143:22.

Searching for 32.197.161.143 in /var/lib/jenkins/.ssh/known_hosts

Searching for 32.197.161.143:22 in /var/lib/jenkins/.ssh/known_hosts

[08/11/26 14:23:22] [SSH] SSH host key matches key in Known Hosts file. Connection will be allowed.

[08/11/26 14:23:22] [SSH] Authentication successful.

[08/11/26 14:23:22] [SSH] The remote user's environment is:

BASH=/usr/bin/bash

BASHOPTS=checkwinsize:cmdhist:complete_fullquote:extquote:force_ignore_globsciiranges:globskipdots:hostcomplete:interactive_comments:patsub

_replacement:progcomp:promptvars:sourcepath

BASHRC_SOURCED=Y

BASH_ALIASES=()

BASH_ARGC=([0]="0")

BASH_ARGV=()

BASH_CMDS=()

BASH_EXECUTION_STRING=set

BASH_LINENO=()

Status

Delete Agent

Configure

Build History

Load Statistics

Log

Build Executor Status

0/2

slave-01

offline

Name ?

slave-01

Description ?

Plain text Preview

Number of executors ?

2

Remote root directory ?

/home/ec2-user/jenkins

Labels ?

java

Save

Apply

Status

Delete Agent

Configure

Build History

Load Statistics

Log

Build Executor Status

0/2

slave-01

offline

Nodes

+ New Node

Configure Monitors

Add, remove, control and monitor the various nodes that Jenkins runs jobs on.

S	Name	Architecture	Clock Difference	Free Disk Space	Free Swap Space	Free Temp Space	Response Time
	Built-In Node	Linux (amd64)	In sync	17.45 GiB	0 B	451.87 MiB	0ms
	slave-01	Linux (amd64)	In sync	5.94 GiB	0 B	456.43 MiB	85ms
Data obtained		4 min 45 sec	4 min 45 sec	4 min 45 sec	4 min 45 sec	4 min 45 sec	4 min 45 sec

Icon:

S

M

L

Legend